

Supplementary material for: Train once, deploy across hospitals? A domain-complete evaluation and an evidence-graded roadmap for cross-site generalization in computational pathology

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ABSTRACT

This document supplements the main article. Sections, figures, tables and equations numbered with the prefix S are in this document; numbers without the prefix refer to the main article.

S1. Site acceptance test: further results

Compact tier Fig. S1 and Table S1 repeat the validation of the main text for the 1,600 compact-tier runs. The compact models differ more between sites, so the development prior is wider and adds less. The test was as safe as at the standard tier, with false acceptance of at most 0.4%, and its gain over local labels alone was confined to small samples.

Cohorts Table S3 breaks down false acceptance and power at a target of 0.85 by cohort. What governs the gain is where the development sites score relative to the target. The median prior center is an AUROC of 0.971 on standard-tier Camelyon17 and 0.897 on MIDOG++, where the test gains most, but 0.869 on the canine cohort and 0.861 on MIDOG 2021, where it gains little. When the development sites score below the target, the prior is pessimistic and costs power: on the compact-tier canine cohort the test accepted no good site at a target of 0.85, against 5–8% for local labels alone. MIDOG 2021 priors rest on only two development sites.

Robust variant The ϵ -contaminated prior replaces the development prior π by $(1-\epsilon)\pi + \epsilon \mathcal{N}(0, 2^2)$ on the logit scale, with $\epsilon = 0.1$; the vague component is centered at an AUROC of 0.5. The local evidence then moves the posterior faster when it contradicts the development sites. Table S2 gives its operating characteristics. We added the variant after the first validation showed false acceptance above 5% in one subgroup (standard-tier Camelyon17 at a target of 0.90 with at most 50 labels), so its validation is not independent of that observation. In that subgroup it reduced false acceptance from 8.7% to 5.4% (25 labels), from 5.7% to 4.5% (50) and from 3.3% to 2.8% (100).

Label planner The planner simulates equal-variance binormal scores at a prevalence of 0.5, with a margin of 0.02 above the target and a target power of 0.8. We drew 150 priors per tier at random and kept those with at least one run 0.02 above a target of 0.85 (114 standard-tier and 44 compact-tier priors; 28 of the 44 are Camelyon17), and compared the planner's predicted acceptance rate of good sites (true AUROC at least 0.02 above a target of 0.85) with the observed rate. At the compact tier the two agreed within 5 percentage points at every sample size. At the standard tier the planner was optimistic by 3 to 7 points (predicted 75% against 68% observed at 100 labels, and 85% against 78% at 200). A practitioner should therefore add a margin to the planned sample size.

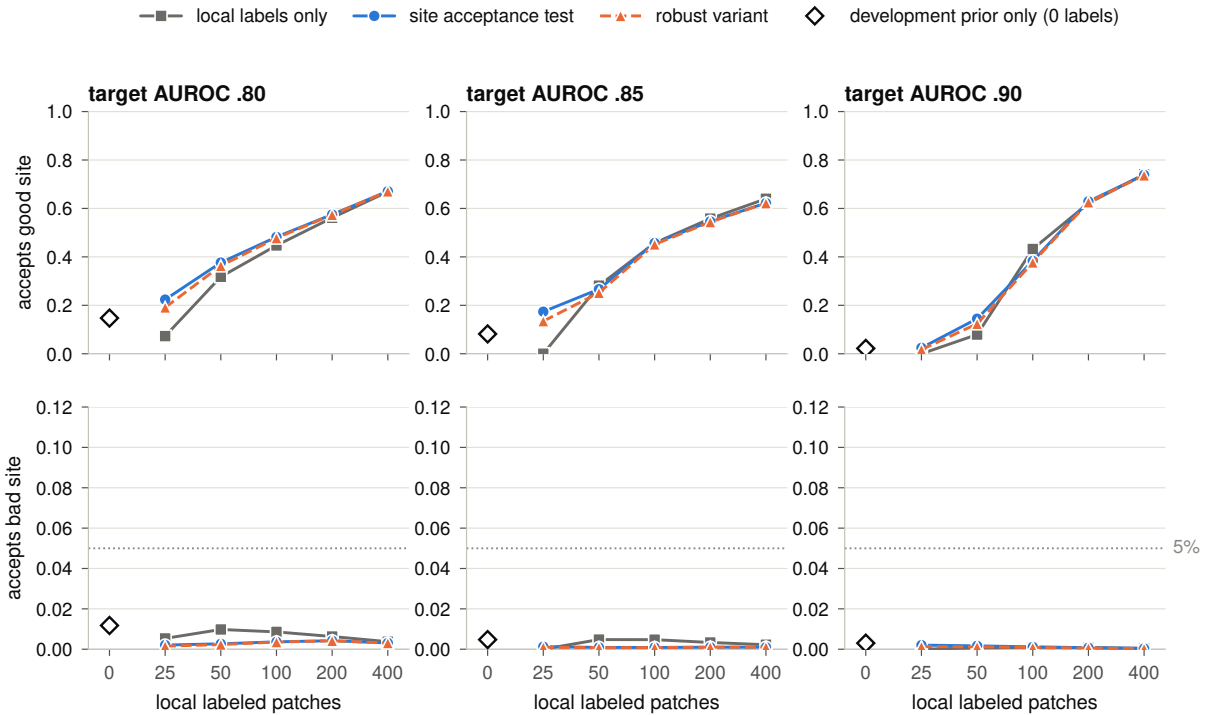
Assumptions The development prior treats the new site as exchangeable with the development sites. In the validation, the prior for site t uses only the other folds' results at their own held-out sites; site t enters those folds only as training data, so its test labels inform neither the prior nor the decision, but the development results are not independent of site t . Local samples are drawn without replacement from the same test patches that define the truth, which for MIDOG 2021 and 400 labels is up to a third of them; this flatters the agreement between estimate and truth somewhat. The decision

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Supplementary material



1600 runs of the compact tier; each held-out site plays the new site, with the prior built from the other folds only. Labeled samples drawn at random from the site's test patches (200 draws per run and size). Accept if $P(\text{AUROC} \geq \text{target}) \geq 0.95$ (local only: one-sided 95% bound $\geq$ target).

Figure S1: Validation of the site acceptance test at the compact tier, as in Fig. 8.

Table S1

Site acceptance test at the compact tier (1,600 runs), as in Table 6.

Rule	m	$T = 0.80$		$T = 0.85$		$T = 0.90$		MAE	cover. (%)
		FA (%)	power (%)	FA (%)	power (%)	FA (%)	power (%)		
Development prior only	0	1.2	15	0.5	8	0.3	2	—	—
Site acceptance test	25	0.2	22	0.1	17	0.2	2	0.042	90
	50	0.3	38	0.1	27	0.2	14	0.035	90
	100	0.4	48	0.1	46	0.1	38	0.027	90
	200	0.4	57	0.1	55	0.1	63	0.021	91
	400	0.3	67	0.1	62	0.0	74	0.015	92
Local labels only	25	0.5	7	0.0	0	0.0	0	0.068	94
	50	1.0	32	0.5	28	0.1	8	0.047	92
	100	0.9	45	0.5	46	0.1	43	0.033	92
	200	0.6	56	0.3	56	0.1	62	0.022	92
	400	0.4	67	0.2	64	0.0	74	0.015	93

threshold is a posterior probability; the reported false-acceptance rates are empirical, not guaranteed. A new site that differs in a way none of the development sites does, for example a new stain protocol, is exactly the case that the local labels must catch. Labeled patches were drawn at random from the site's test patches; in practice patches come in clusters from a few slides, which increases the variance of the local AUROC, so labels should be spread over many slides. The test concerns patch-level AUROC, not a clinical endpoint.

Table S2

Robust variant ($\epsilon = 0.1$) at the standard tier (top) and the compact tier (bottom), as in Table 6. The variant was added after the first validation and is therefore not independently validated.

Rule	m	$T = 0.80$		$T = 0.85$		$T = 0.90$		MAE	cover. (%)
		FA (%)	power (%)	FA (%)	power (%)	FA (%)	power (%)		
Robust variant	25	1.4	58	0.6	28	0.2	14	0.029	93
	50	1.7	72	0.8	47	0.3	27	0.024	92
	100	1.2	83	0.7	62	0.2	43	0.019	92
	200	0.8	89	0.6	71	0.2	51	0.015	92
	400	0.4	93	0.3	79	0.1	60	0.011	93
Robust variant	25	0.2	19	0.1	13	0.1	2	0.043	91
	50	0.2	36	0.1	25	0.1	12	0.035	91
	100	0.4	48	0.1	45	0.1	38	0.027	91
	200	0.4	57	0.1	54	0.1	63	0.021	91
	400	0.3	67	0.1	62	0.0	74	0.015	92

Table S3

False acceptance (%) / power (%) by cohort at a target of 0.85, with 50 and 100 local labels, for the site acceptance test, its robust variant and local labels alone. Bad: number of runs whose true site AUROC is below 0.85.

Tier	Cohort	bad	Site 50	Site 100	Robust 50	Robust 100	Local 50	Local 100
S	Camelyon17	14	2.5 / 87	0.9 / 95	1.9 / 81	0.7 / 93	0.3 / 60	0.2 / 88
	Canine cSCC	99	1.5 / 27	1.2 / 45	1.4 / 26	1.1 / 44	0.9 / 26	0.7 / 45
	MIDOG 2021	62	0.0 / 5	0.2 / 14	0.0 / 4	0.2 / 13	0.4 / 4	0.7 / 11
	MIDOG++	73	0.7 / 50	0.6 / 65	0.6 / 46	0.5 / 63	0.3 / 15	0.3 / 34
C	Camelyon17	114	0.8 / 49	0.7 / 80	0.7 / 46	0.6 / 79	0.4 / 49	0.4 / 76
	Canine cSCC	376	0.0 / 0	0.0 / 0	0.0 / 0	0.0 / 0	0.4 / 5	0.3 / 8
	MIDOG 2021	173	0.0 / 3	0.0 / 10	0.0 / 2	0.0 / 9	0.3 / 3	0.3 / 9
	MIDOG++	387	0.0 / 3	0.0 / 9	0.0 / 2	0.0 / 9	0.7 / 6	0.7 / 15

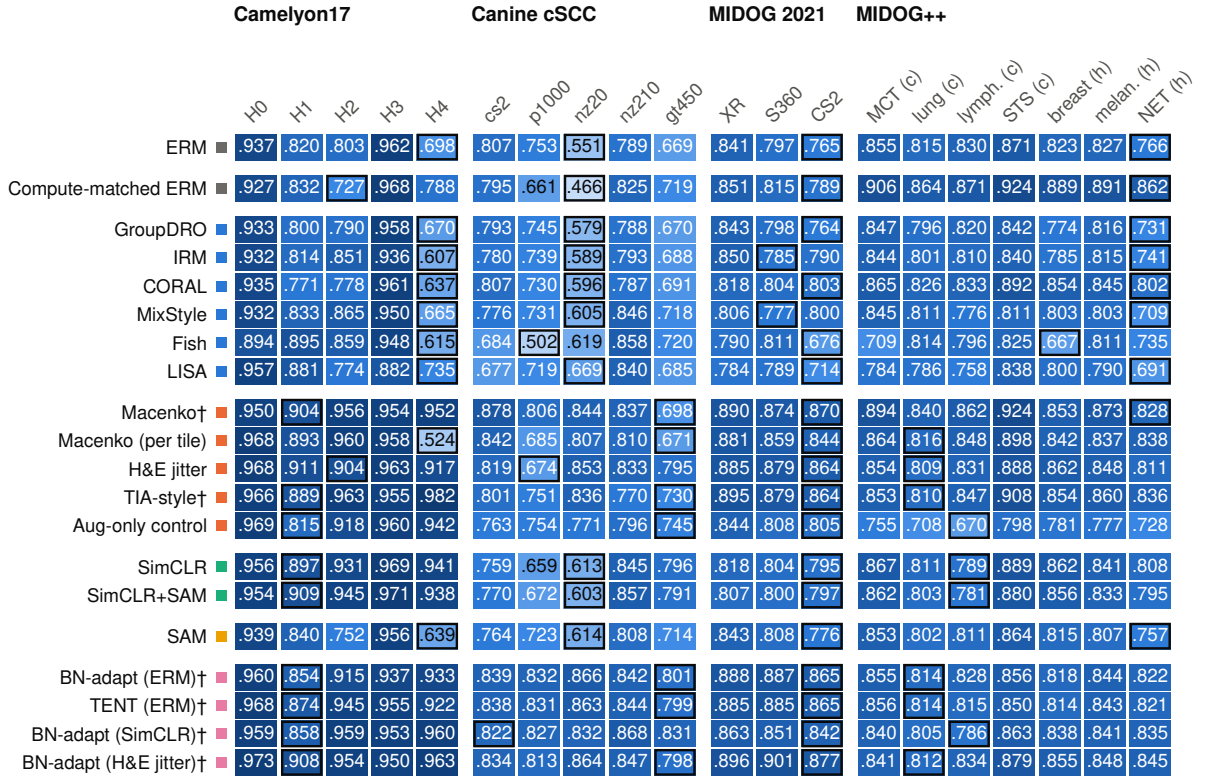
S2. Compact-tier results

This section gives the compact-tier counterparts of the main-text figures and tables. Table S4 lists worst-domain and mean AUROC, Fig. S2 the per-domain results, Fig. S3 the variance decomposition, Fig. S4 the power curves and Fig. S5 the rankings. Figure S6 shows the individual seeds of three compact-tier arms next to the best standard-tier arm.

Table S4

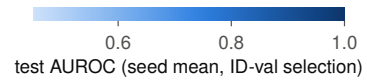
Compact tier: worst-domain AUROC (rank among the 20 arms) and mean AUROC over held-out domains, ID-validation selection, five seeds. Bold: best worst-domain AUROC in the cohort. [†]Transductive.

Arm	Camelyon17 ($K = 5$)		Canine cSCC ($K = 5$)		MIDOG 2021 ($K = 3$)		MIDOG++ ($K = 7$)	
	worst (rank)	mean	worst (rank)	mean	worst (rank)	mean	worst (rank)	mean
<i>Reference and controls</i>								
ERM	0.698 (13)	0.844	0.551 (18)	0.714	0.765 (17)	0.801	0.766 (13)	0.827
Compute-matched ERM	0.727 (12)	0.848	0.466 (20)	0.693	0.789 (13)	0.818	0.862 (1)	0.887
<i>Domain-aware training</i>								
GroupDRO	0.670 (14)	0.830	0.579 (17)	0.715	0.764 (18)	0.802	0.731 (16)	0.804
IRM	0.607 (19)	0.828	0.589 (16)	0.718	0.785 (14)	0.809	0.741 (15)	0.805
CORAL	0.637 (17)	0.816	0.596 (15)	0.723	0.803 (10)	0.808	0.802 (9)	0.845
MixStyle	0.665 (15)	0.849	0.605 (13)	0.735	0.777 (15)	0.794	0.709 (17)	0.794
Fish	0.615 (18)	0.842	0.502 (19)	0.676	0.676 (20)	0.759	0.667 (20)	0.765
LISA	0.735 (11)	0.846	0.669 (10)	0.718	0.714 (19)	0.762	0.691 (18)	0.778
<i>Stain normalization and color</i>								
Macenko [†]	0.904 (4)	0.943	0.698 (7)	0.813	0.870 (2)	0.878	0.828 (2)	0.868
Macenko (per tile)	0.524 (20)	0.861	0.671 (9)	0.763	0.844 (7)	0.861	0.816 (3)	0.849
H&E jitter	0.904 (3)	0.933	0.674 (8)	0.795	0.864 (5)	0.876	0.809 (8)	0.843
TIA-style [†]	0.889 (6)	0.951	0.730 (6)	0.777	0.864 (6)	0.879	0.810 (7)	0.853
Aug-only control	0.815 (10)	0.921	0.745 (5)	0.766	0.805 (9)	0.819	0.670 (19)	0.745
<i>Self-supervision</i>								
SimCLR	0.897 (5)	0.939	0.613 (12)	0.734	0.795 (12)	0.806	0.789 (10)	0.838
SimCLR+SAM	0.909 (1)	0.944	0.603 (14)	0.738	0.797 (11)	0.801	0.781 (12)	0.830
<i>Sharpness-aware</i>								
SAM	0.639 (16)	0.825	0.614 (11)	0.725	0.776 (16)	0.809	0.757 (14)	0.815
<i>Test-time adaptation</i>								
BN-adapt (ERM) [†]	0.854 (9)	0.920	0.801 (2)	0.836	0.865 (3)	0.880	0.814 (4)	0.834
TENT (ERM) [†]	0.874 (7)	0.933	0.799 (3)	0.835	0.865 (4)	0.878	0.814 (5)	0.830
BN-adapt (SimCLR) [†]	0.858 (8)	0.938	0.822 (1)	0.836	0.842 (8)	0.852	0.786 (11)	0.830
BN-adapt (H&E jitter) [†]	0.908 (2)	0.950	0.798 (4)	0.831	0.877 (1)	0.891	0.812 (6)	0.845



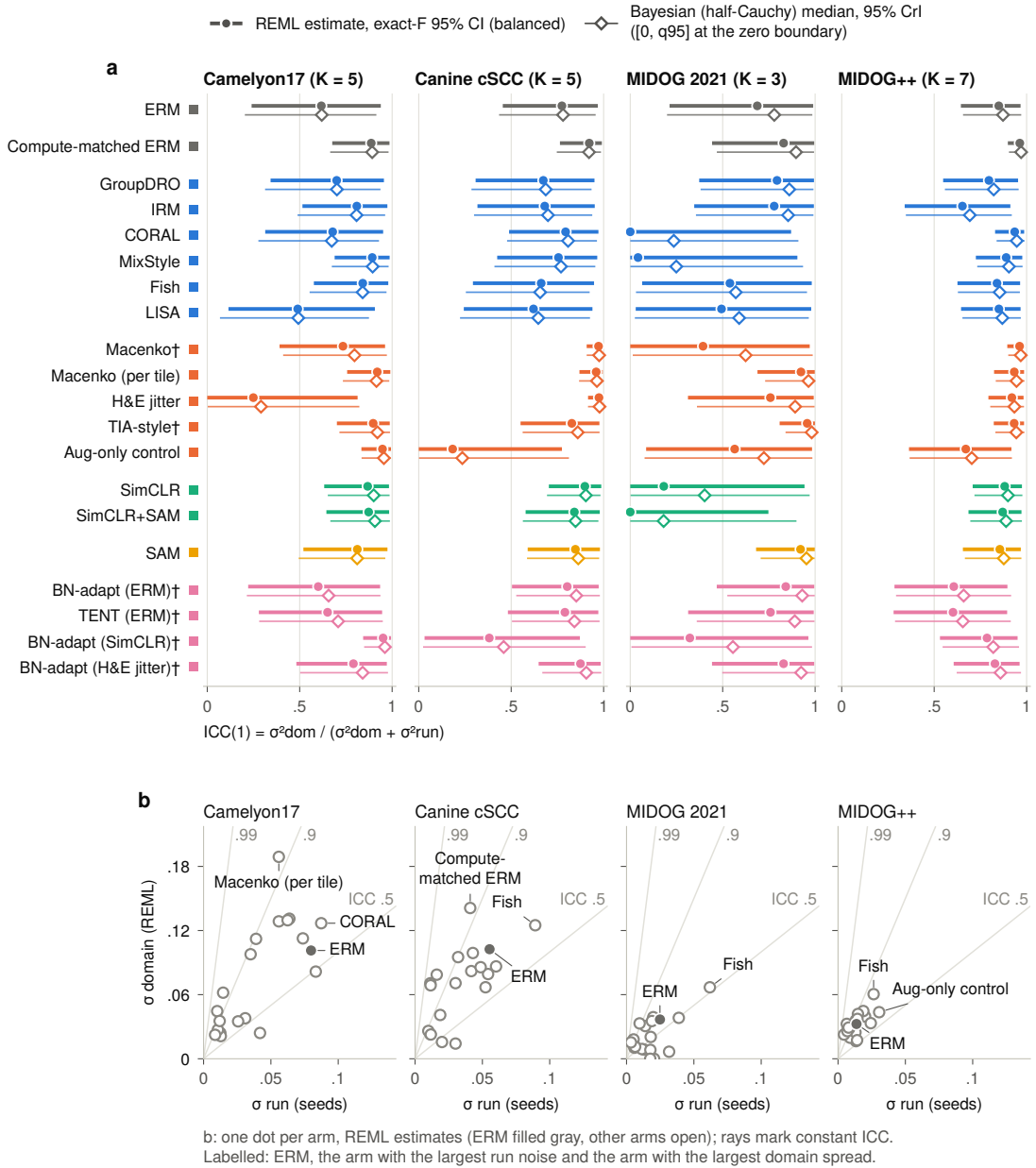
Outlined: worst domain of the arm (rows with every domain present).

■ Reference ■ Stain & color ■ Sharpness
■ DG objectives ■ Self-supervision ■ Test-time adaptation



† Transductive: uses the unlabeled images of the test domain at prediction time.

Figure S2: Test AUROC of every compact-tier arm at every held-out domain (mean over five seeds; ID-validation selection). Outlined: each arm's worst domain. † Transductive.



† Transductive: uses the unlabeled images of the test domain at prediction time.

Figure S3: Variance decomposition at the compact tier, as in Fig. 3. † Transductive.

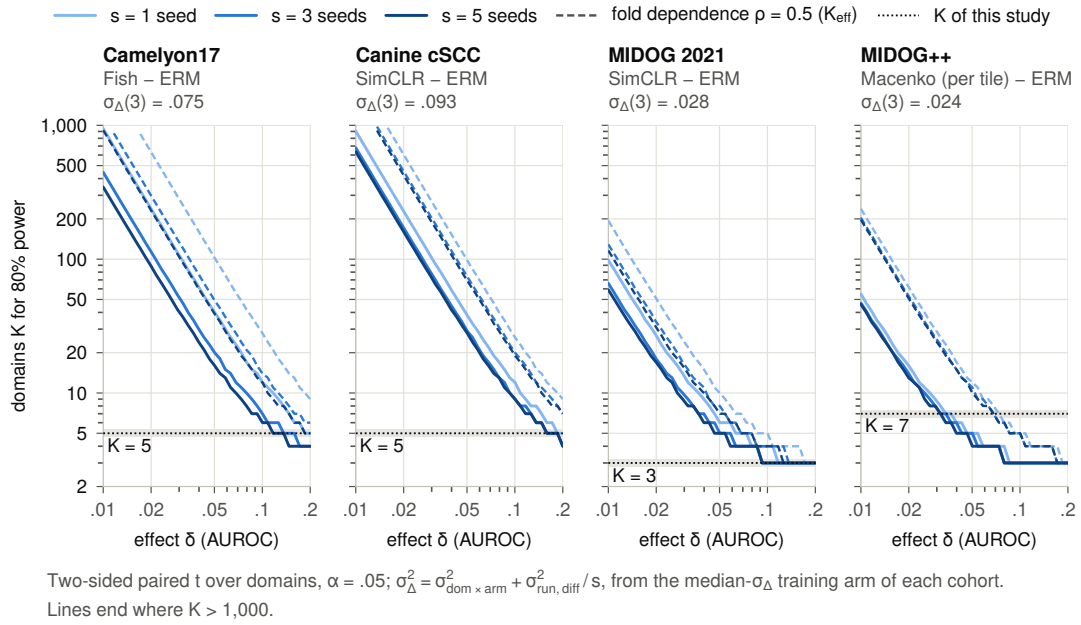
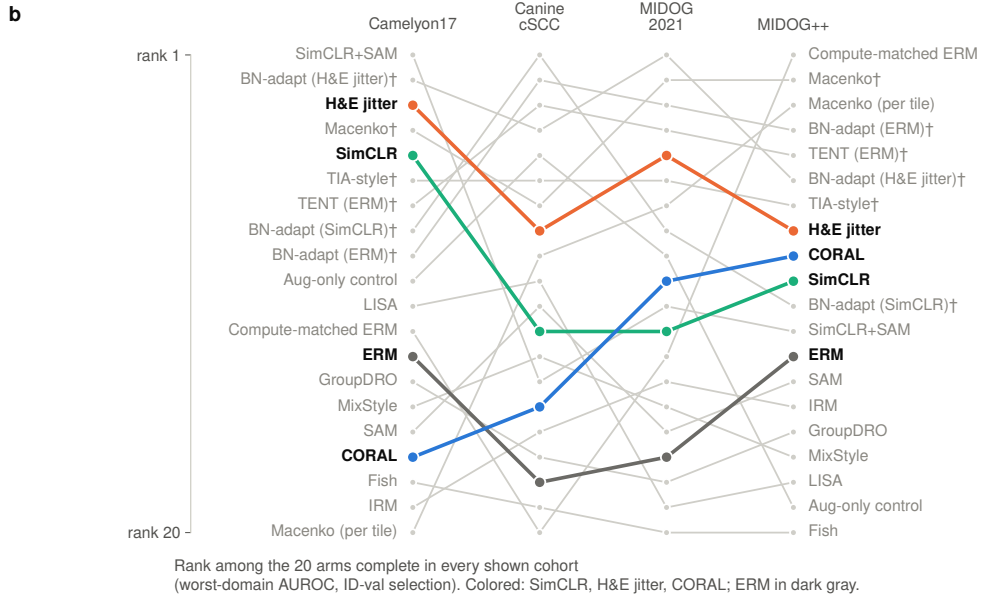
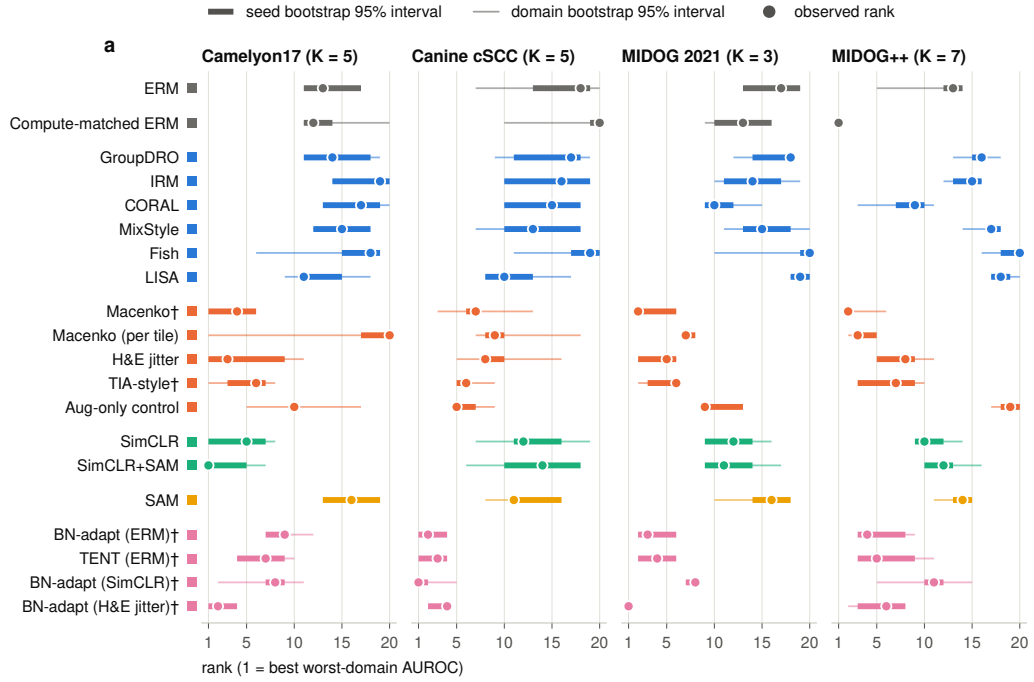


Figure S4: Domains needed for 80% power at the compact tier, as in Fig. 4.



† Transductive: uses the unlabeled images of the test domain at prediction time.

Figure S5: Rankings at the compact tier, as in Fig. 5. †Transductive.

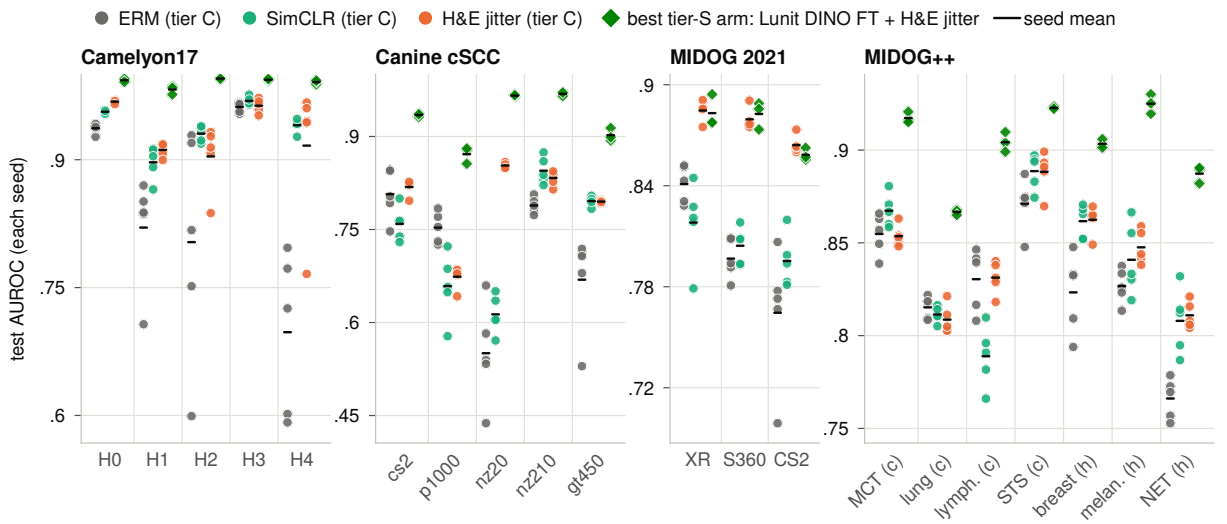
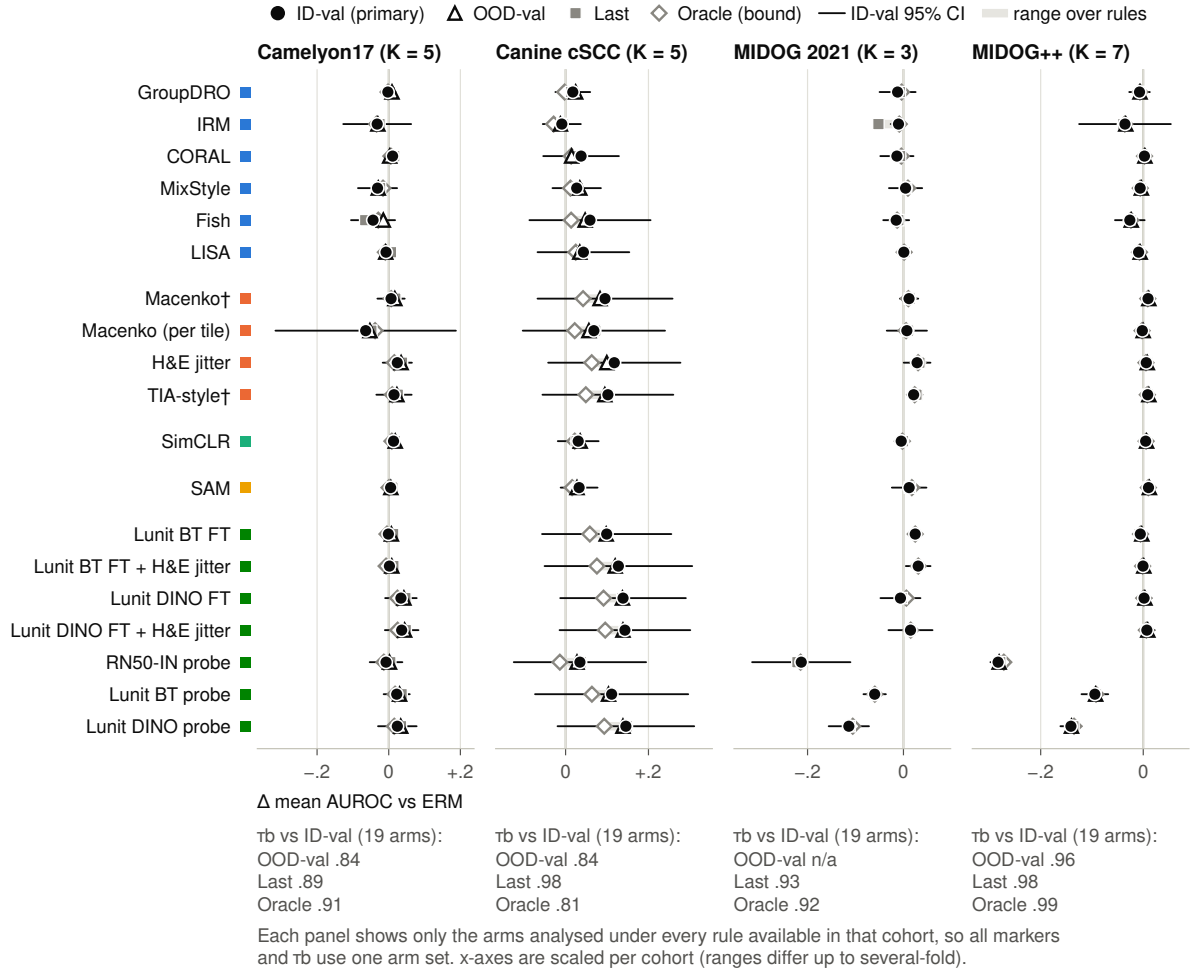


Figure S6: Individual seeds at every held-out domain for compact-tier ERM, SimCLR and H&E jitter, and for the best standard-tier arm (fine-tuned Lunit DINO with H&E jitter, the arm with the highest mean worst-domain AUROC over the four cohorts, 0.895). Bars mark seed means. The y-axes are scaled per cohort.

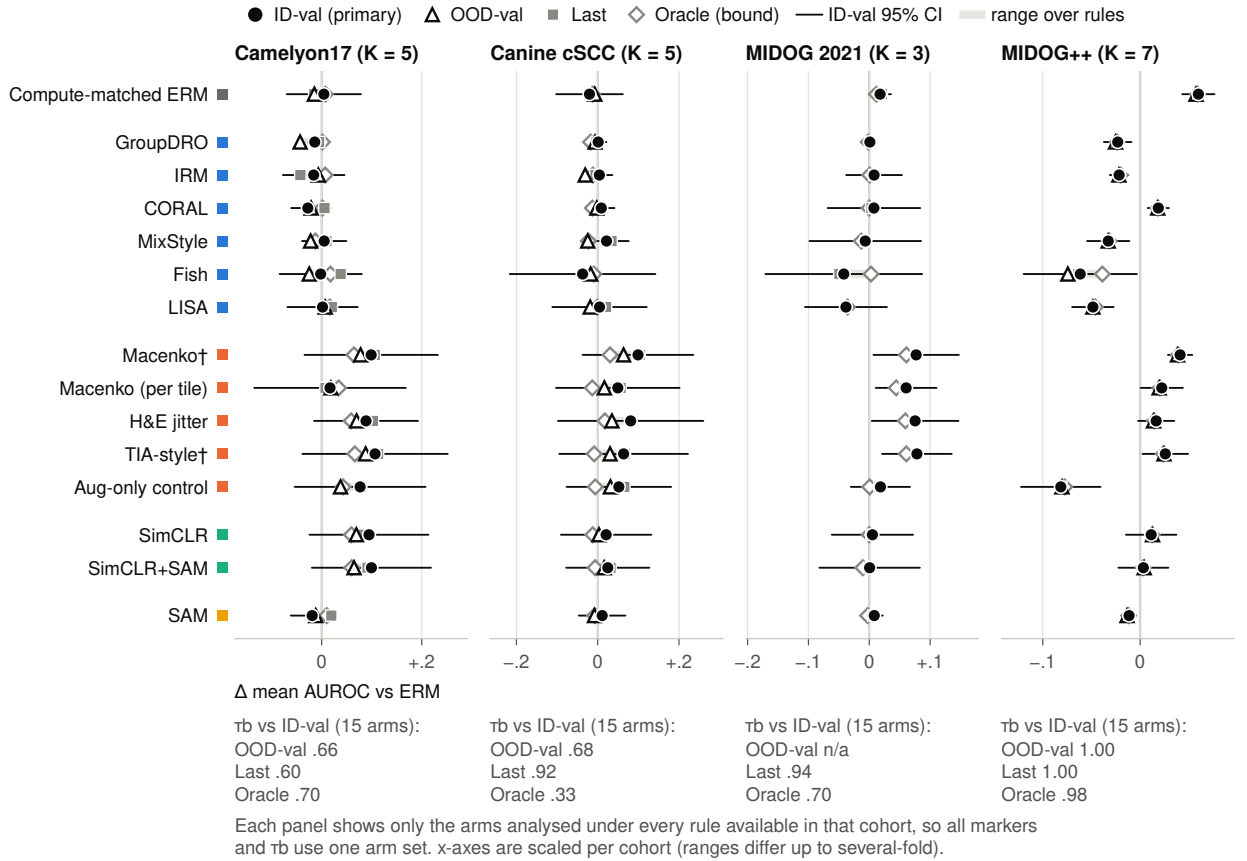
S3. Model-selection rules

Figures S7 and S8 show each arm's mean difference from ERM under the four selection rules. All rules use the same ten checkpoints of the same runs (Section 3.1), so the differences come from selection alone. The τ_b values printed in the figures compare the rankings of these mean differences; those of Table 5 compare worst-domain rankings. At the standard tier the rules agree closely. At the compact tier the OOD-validation and oracle rules reorder the canine and Camelyon17 arms considerably.



† Transductive: uses the unlabeled images of the test domain at prediction time.

Figure S7: Standard tier: mean AUROC difference from ERM under ID-validation (primary, with 95% interval), OOD-validation, last-checkpoint and oracle selection. Each panel shows the arms analyzed under every rule available in the cohort. † Transductive.

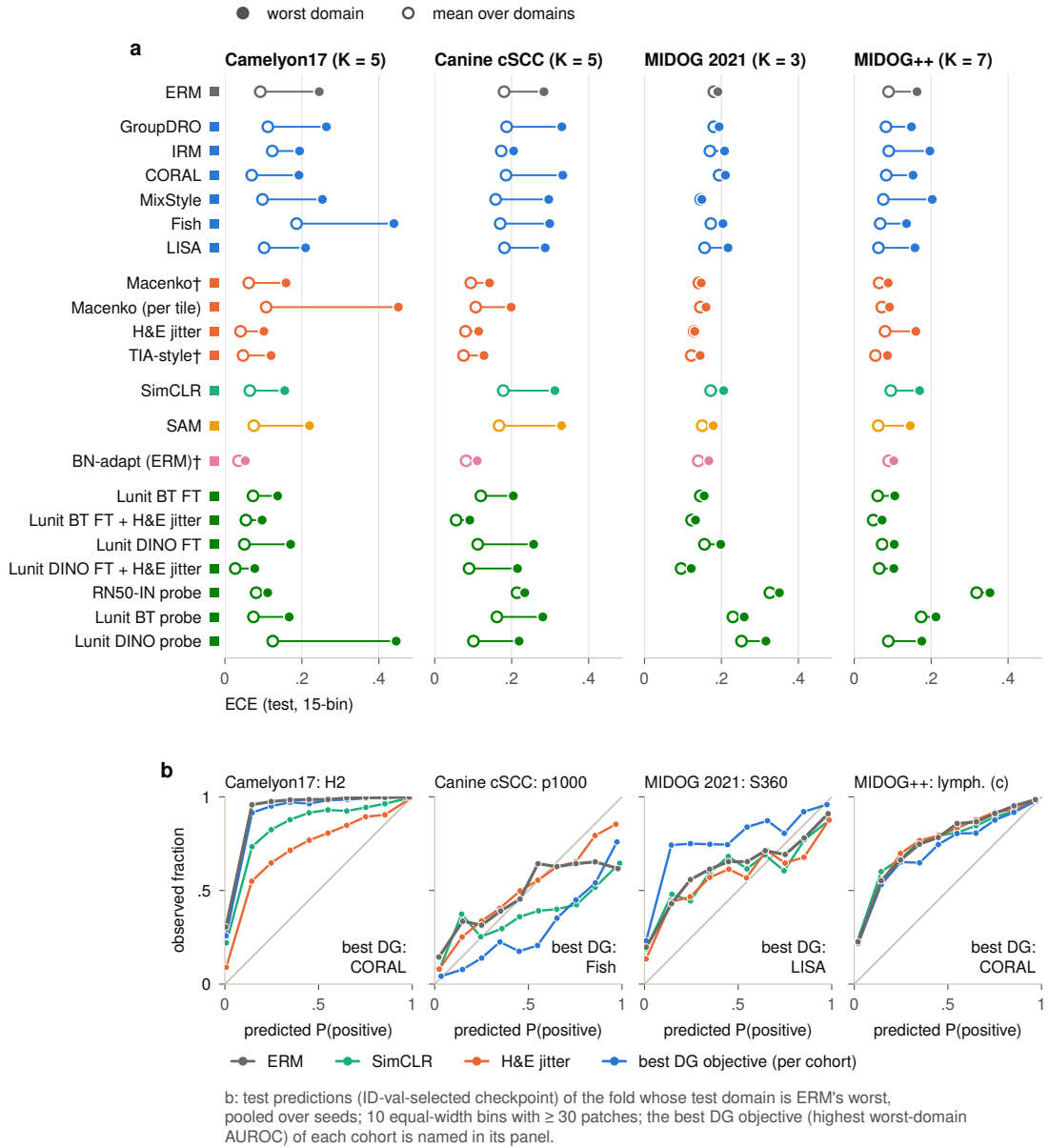


† Transductive: uses the unlabeled images of the test domain at prediction time.

Figure S8: Compact tier: mean AUROC difference from ERM under the four selection rules, as in Fig. S7. †Transductive.

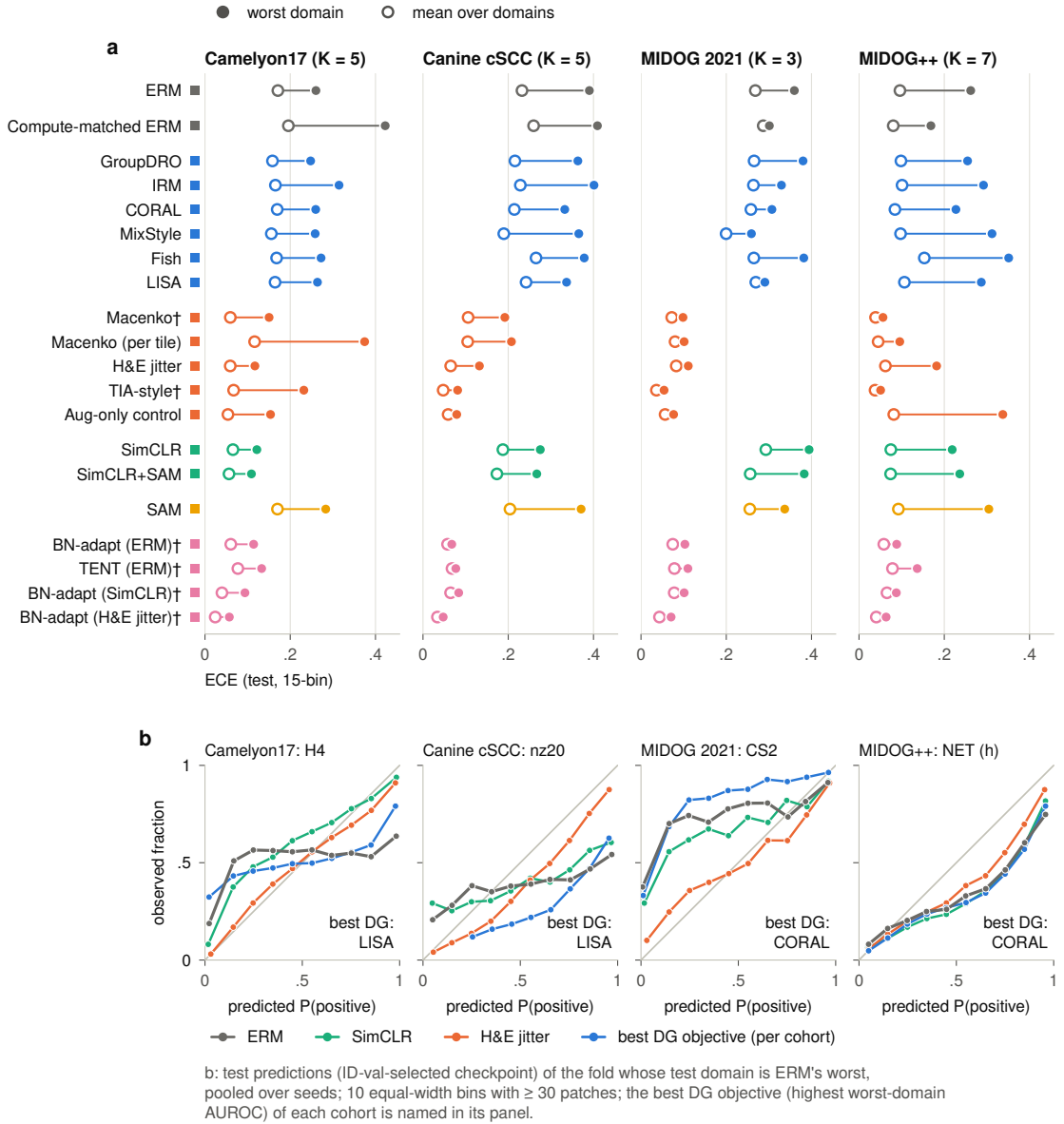
S4. Calibration

Figures S9 and S10 give the expected calibration error (Eq. 4) of every arm, averaged over domains and at its worst domain, and reliability diagrams for the fold in which ERM performs worst. At the standard tier, one of the two fine-tuned foundation models with H&E jitter is the best calibrated arm on every cohort. ERM is poorly calibrated on the canine cohort and MIDOG 2021 (mean ECE 0.180 on both). Calibration varies between domains as much as discrimination does: for most arms the worst-domain ECE is well above the mean.



† Transductive: uses the unlabeled images of the test domain at prediction time.

Figure S9: Calibration at the standard tier. (a) Test ECE (15 equal-width bins) at the worst domain (filled) and averaged over domains (open). (b) Reliability diagrams for the fold whose test domain is ERM's worst, pooled over seeds. †Transductive.



† Transductive: uses the unlabeled images of the test domain at prediction time.

Figure S10: Calibration at the compact tier, as in Fig. S9. †Transductive.

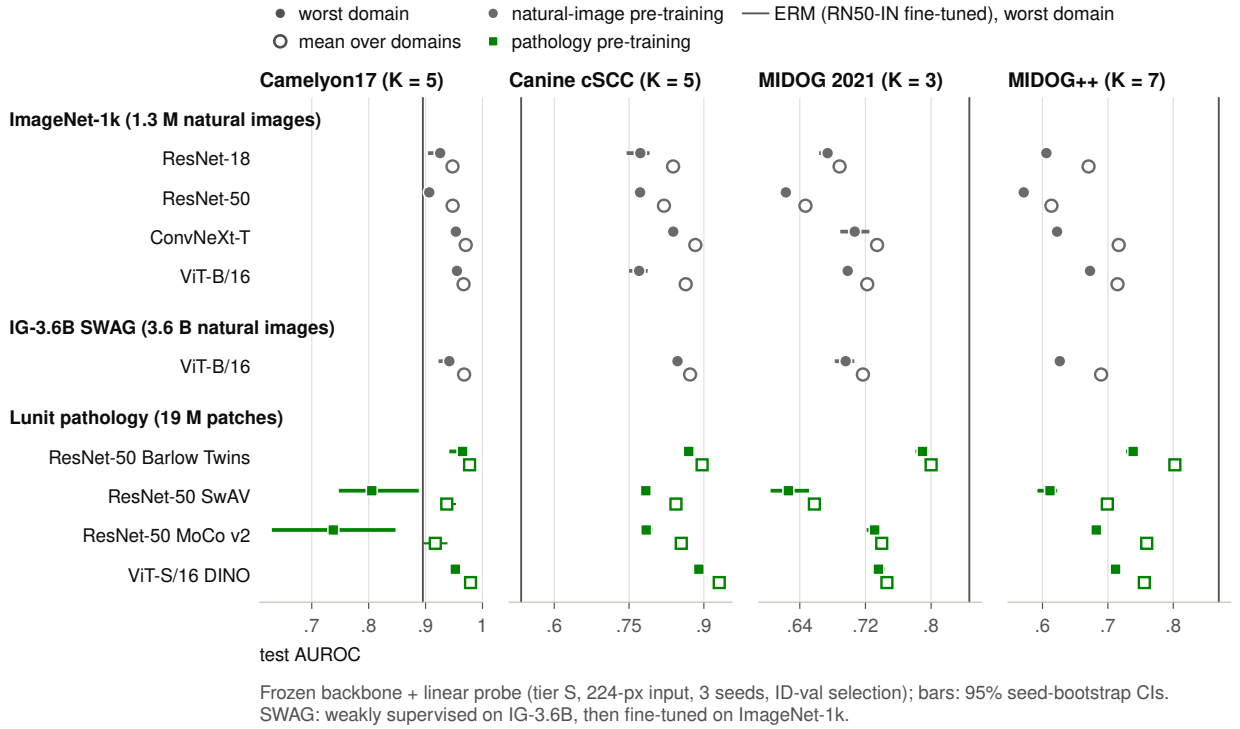


Figure S11: Frozen backbones with a linear probe (standard tier, 224 px input, three seeds, ID-validation selection): worst-domain (filled, with seed-bootstrap 95% interval) and mean AUROC (open). Vertical line: fine-tuned ImageNet ResNet-50 ERM, worst domain.

S5. Representation ladder

Figure S11 compares nine frozen backbones under one linear-probe protocol: four ImageNet-1k models (ResNet-18, ResNet-50, ConvNeXt-T (Liu et al., 2022), ViT-B/16), a ViT-B/16 pre-trained with weak supervision on 3.6 billion images and fine-tuned on ImageNet-1k (Singh et al., 2022), and four Lunit pathology models (Kang et al., 2023). The Lunit Barlow Twins ResNet-50 and DINO ViT-S/16 are the best frozen encoders on the canine cohort and, by mean AUROC, on Camelyon17; by worst-domain AUROC on Camelyon17 the DINO ViT is fourth, behind Barlow Twins and two ImageNet models. On the mitosis cohorts every frozen probe is well below fine-tuned ERM, and the Lunit SwAV ResNet-50 is below the ImageNet ResNet-50 averaged over the four cohorts. Within this probe-only ladder, the two mitosis cohorts rank the backbones almost identically ($\tau_b = 0.83$), while the two tumor cohorts barely agree ($\tau_b = 0.17$).

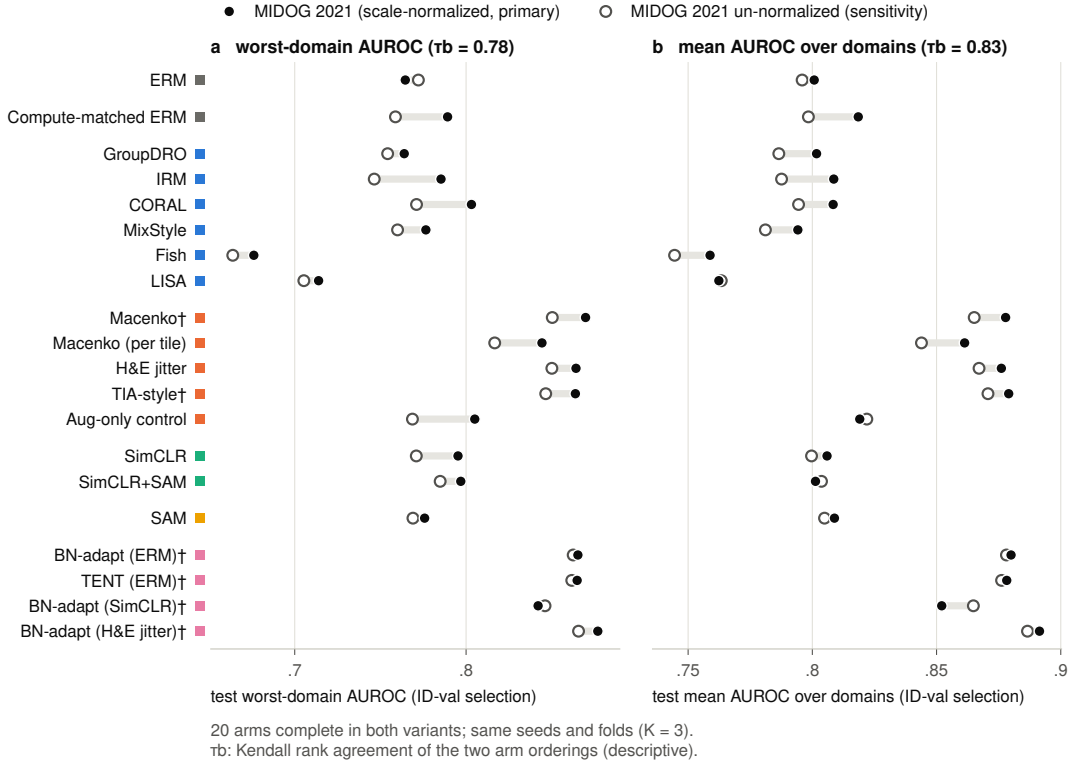


Figure S12: MIDOG 2021 with and without scale normalization (compact tier, 20 arms, same seeds and folds): (a) worst-domain and (b) mean AUROC. † Transductive.

S6. Sensitivity analyses

Scale normalization of MIDOG 2021 Figure S12 compares the compact-tier results on the scale-normalized (primary) and the un-normalized MIDOG 2021 cohort.

PatchCamelyon Table S5 gives the PatchCamelyon test AUROC of the Camelyon17 models at the ID-validation-selected checkpoint, averaged over seeds, with its mean and minimum over the five Camelyon17 folds. The score is never used for selection. Test-time adaptation is not evaluated on PatchCamelyon.

Hyperparameters Figures S13 and S14 show how test AUROC of the seed-0 tuning runs varies across each grid. Test AUROC is shown only to display this sensitivity; selection used ID-validation AUROC (Section S8).

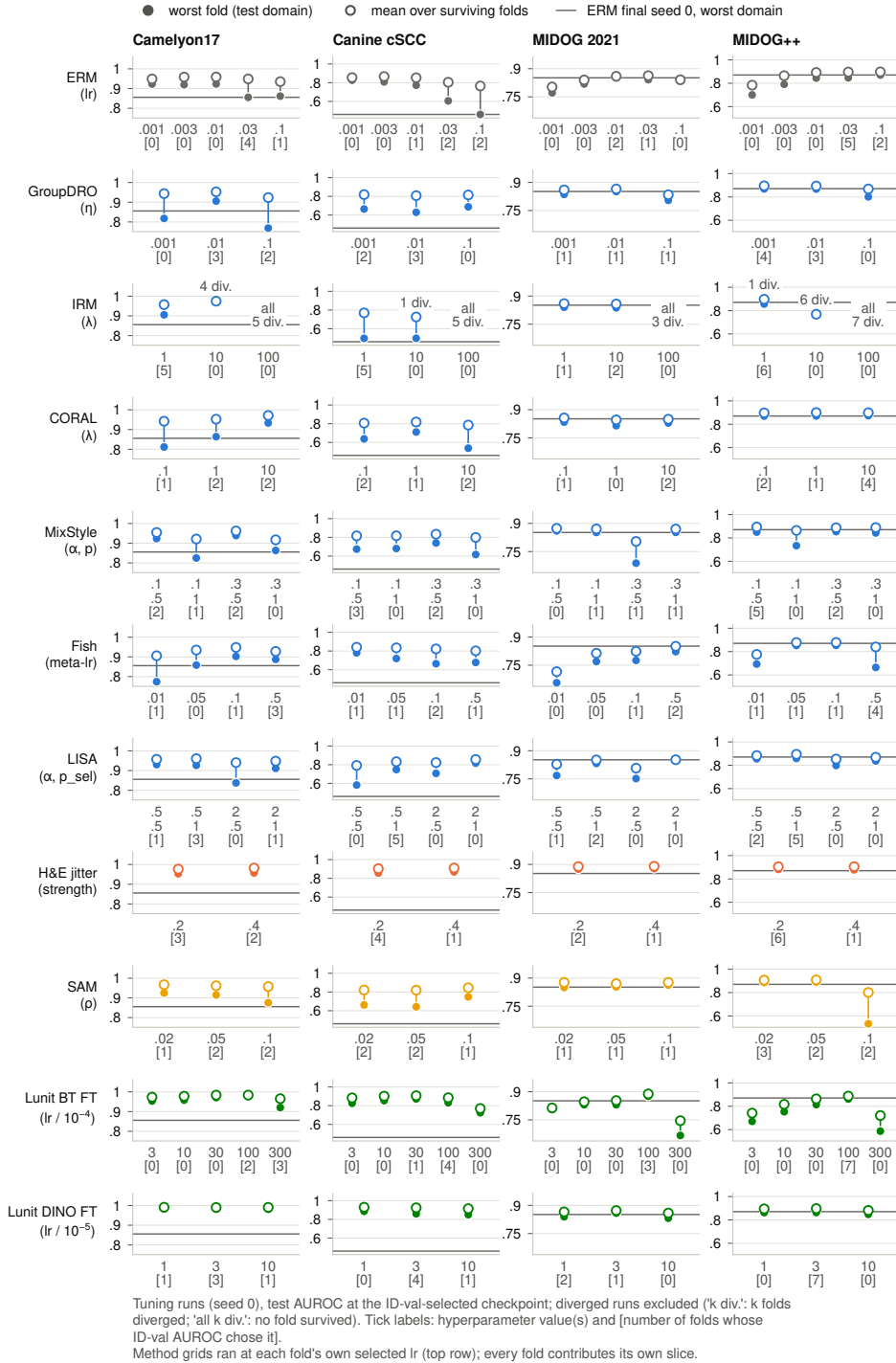


Figure S13: Standard-tier tuning runs (seed 0): test AUROC at the ID-validation-selected checkpoint for each grid value, worst fold (filled) and mean over folds (open). Brackets: number of folds whose ID-validation AUROC selected the value. 'k div.': k folds diverged. Gray line: worst domain of final ERM, seed 0.

Table S5

PatchCamelyon test AUROC of the Camelyon17 models: mean and minimum over the five folds of the seed-mean AUROC.

†Transductive.

Arm	Compact tier		Standard tier	
	mean	min over folds	mean	min over folds
ERM	0.698	0.641	0.772	0.665
Compute-matched ERM	0.652	0.542	—	—
GroupDRO	0.697	0.634	0.791	0.738
IRM	0.716	0.675	0.753	0.669
CORAL	0.691	0.621	0.772	0.707
MixStyle	0.739	0.703	0.780	0.702
Fish	0.647	0.581	0.729	0.618
LISA	0.672	0.576	0.756	0.631
Macenko [†]	0.765	0.677	0.752	0.634
Macenko (per tile)	0.756	0.686	0.837	0.784
H&E jitter	0.768	0.664	0.843	0.801
TIA-style [†]	0.802	0.730	0.858	0.838
Aug-only control	0.777	0.725	—	—
SimCLR	0.743	0.701	0.753	0.656
SimCLR+SAM	0.761	0.724	—	—
SAM	0.690	0.544	0.780	0.684
Lunit BT FT	—	—	0.838	0.791
Lunit BT FT + H&E jitter	—	—	0.850	0.781
Lunit DINO FT	—	—	0.905	0.874
Lunit DINO FT + H&E jitter	—	—	0.908	0.895
RN50-IN probe	—	—	0.820	0.806
Lunit BT probe	—	—	0.889	0.866
Lunit DINO probe	—	—	0.886	0.867

Supplementary material

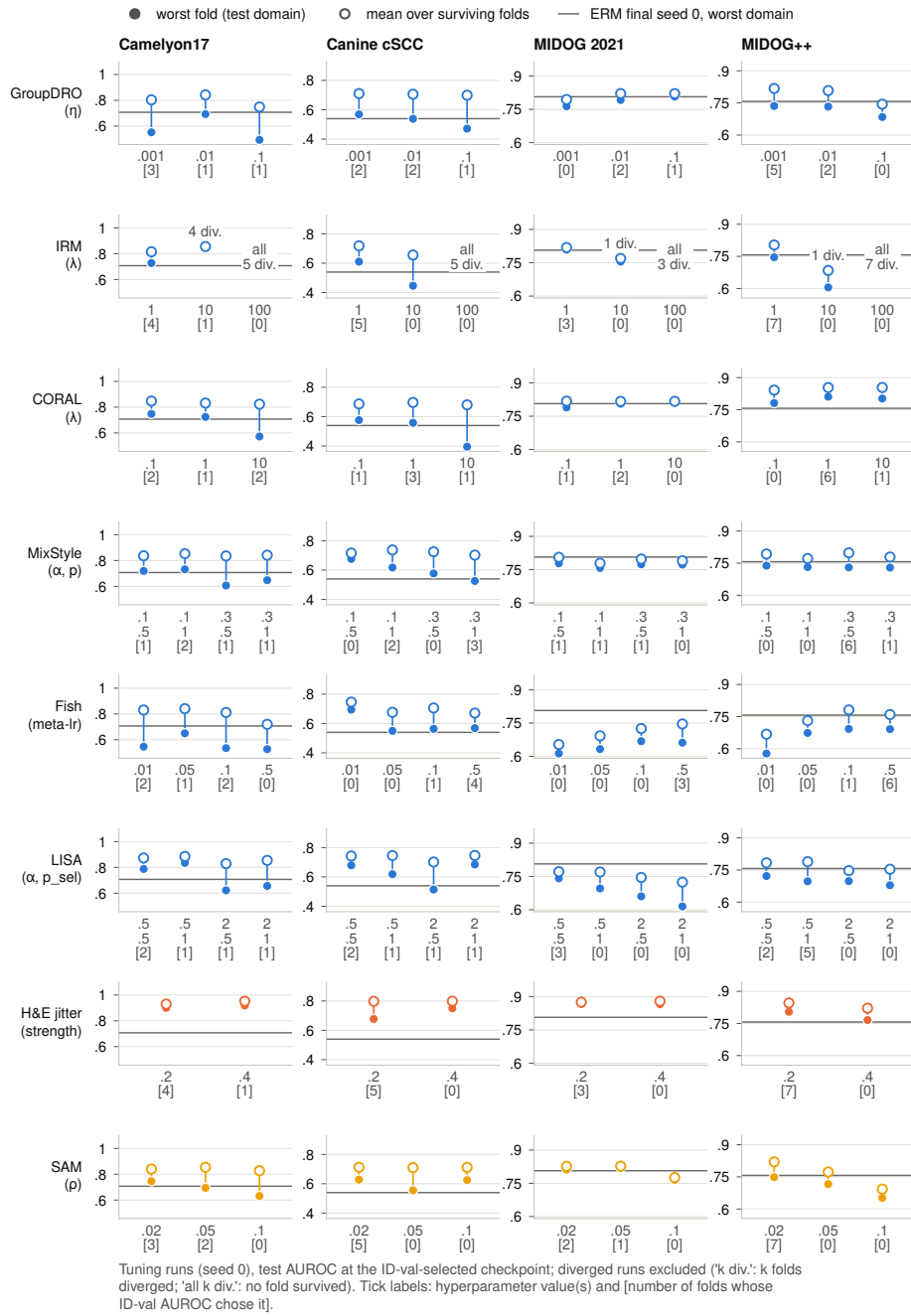


Figure S14: Compact-tier tuning runs (seed 0), as in Fig. S13.

Table S6

Response shape: number of arms with negative Oldham's correlation r , median r , number of arms with Pitman–Morgan $p < 0.05$, and the smallest exact sign-flip p over arms.

Tier	Cohort	K	$r < 0$	median r	Pitman–Morgan $p < 0.05$	smallest exact p
S	Camelyon17	5	15/20	-0.36	7	0.062
	Canine cSCC	5	20/20	-0.91	14	0.062
	MIDOG 2021	3	10/20	-0.01	0	0.250
	MIDOG++	7	4/20	+0.13	5	0.078
C	Camelyon17	5	12/19	-0.52	7	0.125
	Canine cSCC	5	17/19	-0.73	6	0.062
	MIDOG 2021	3	16/19	-0.94	6	0.250
	MIDOG++	7	10/19	-0.03	0	0.281

S7. Response shape

Table S6 summarizes Oldham's correlation r between each arm's per-domain gain over ERM and the per-domain mean of the two (Section S11.3). A negative r means that the arm gains most where both arms do worst, so it narrows the spread between domains. On the tumor cohorts at both tiers and on MIDOG 2021 at the compact tier, most arms have a negative r ; on the canine cohort at the standard tier, all 20 do. The Pitman–Morgan test is significant for 14 of them. The exact sign-flip test cannot reach $p < 0.05$ with five or fewer domains, and on MIDOG++ its smallest p is 0.078 at the standard tier. We therefore read the pattern as consistent but not established.

S8. Hyperparameter tuning and diverged runs

Protocol Hyperparameters are tuned with seed 0 on the grids of Table 2, separately for every fold (Section 3.1). Each fold keeps the configuration with the highest ID-validation AUROC at the ID-selected checkpoint of its own tuning run. Diverged runs are excluded unless every configuration of the fold diverged. In that case the highest ID-validation AUROC among the diverged runs is used, and ties are broken by a fixed ordering of the configurations. Test data are never used. The tuning records are kept apart from the final runs. The compact tier has 26 configurations, which gives 520 tuning runs on the 20 folds of the four primary cohorts and 78 on the three folds of the un-normalized MIDOG 2021 sensitivity cohort. The standard tier has 160 learning-rate runs (five ResNet-50 and three ViT rates on 20 folds), 100 Barlow Twins learning-rate runs (five rates on 20 folds; Section S8.1) and 520 method-grid runs. All 1,378 tuning records are present, and every arm has a selection in every fold. The following arms have no tuning grid: compact-tier ERM (learning rate 0.02), augmentation-only, compute-matched ERM, SimCLR, the Macenko family (per-slide Macenko, per-tile Macenko and TIA-style), BN-adapt, TENT and the frozen probes. Their values are fixed as in Table 2, except that every standard-tier ResNet-50 arm takes its fold’s tuned learning rate (Section S8.1). Further arms inherit tuned values: SimCLR+SAM inherits ρ from SAM, Barlow Twins with H&E jitter the fold’s Barlow Twins ERM learning rate, and the foundation-model H&E-jitter arms s from standard-tier H&E jitter.

Grid edges A selected value is *at the edge* when it is the smallest or largest value of its grid axis. We flag an axis value when it is an edge value and is selected on more than half (at least 11) of the 20 primary folds, since the optimum may then lie outside the grid. On two-value axes every value is an edge, so a flag there only shows that one value dominates.

S8.1. Standard-tier learning rate

At the standard tier the base learning rate is selected first, on ImageNet ERM, and is shared by every ResNet-50 arm of the fold. The method grids are then run at that rate. Every stage-2 tuning run used its fold’s selected rate. The ResNet-50 grid was extended twice because the optimum sat on its upper edge (Tables S8 and S9). On the initial grid $\{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$, 10^{-2} was selected on 19 of 20 folds. After 3×10^{-2} was added, it was selected on 16 of 20 folds. Its ID-validation AUROC exceeded that of 10^{-2} on the same 16 folds, by a median of +0.0035 (range -0.018 to $+0.046$). The per-cohort medians ranged from -0.0019 (MIDOG 2021) to $+0.0084$ (MIDOG++). Since 3×10^{-2} was again an edge on most folds, 10^{-1} was added. The ID-validation AUROC of 10^{-1} minus that of 3×10^{-2} had a median of -0.0023 (range -0.022 to $+0.013$) and was positive on only 5 of 20 folds. The final picks are 3×10^{-2} on 12 folds, 10^{-1} on 5 and 10^{-2} on 3. The best ID-validation AUROC of a fold rose by a median of 0.0035 with the first extension and did not change in the median with the second (15 of 20 folds unchanged). The five folds that select 10^{-1} still sit on the upper edge. Their gain over 3×10^{-2} is 0.0012–0.0133 (median 0.0052), and the grid was not extended further. The ViT learning rate was selected from $\{10^{-5}, 3 \times 10^{-5}, 10^{-4}\}$ without extension: 3×10^{-5} on 15 folds, 10^{-5} on 3 and 10^{-4} on 2. None of the 160 learning-rate runs diverged. The per-fold picks under the shorter grids in Table S9 are recomputed from the final records, by restricting each fold to the rates of the shorter grid.

Barlow Twins learning rate The Barlow Twins fine-tunes first inherited the fold’s ImageNet ResNet-50 rate. On 17 of 20 folds the ImageNet rate is 3×10^{-2} or 10^{-1} , and with it training failed on several folds: the ID-validation AUROC was much lower than at smaller rates, and single runs reached an off-site AUROC of only 0.49 on MIDOG++. We therefore gave them their own per-fold rate, selected on Barlow Twins ERM (seed 0) from $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$ and shared with Barlow Twins plus H&E jitter. None of the 100 runs diverged. The picks are 10^{-2} on 16 folds, 3×10^{-2} on 3 (Camelyon17 folds 0–2) and 3×10^{-3} on 1 (canine fold 4). The optimum is interior on every fold. The records with the inherited rate are kept apart from the final runs. Table S7 compares them with the final records. On the 6 folds where both rates agree, the records are identical. The tuned rate raises the mean off-site AUROC most on MIDOG++, from 0.715 to 0.891 for ERM and from 0.687 to 0.896 with H&E jitter. A pretrained backbone that inherits the learning rate of a different initialization can therefore look much worse than it is.

S8.2. Method grids

Tables S10 and S11 count how often each grid value is selected. At the compact tier, the flagged edges are the smallest values of IRMv1’s λ (19 of 20 folds) and SAM’s ρ (17), the largest Fish step $\epsilon = 0.5$ (13), and the dominant values of the two-value axes of H&E jitter, LISA and MixStyle. DeepCORAL and GroupDRO are not flagged. At the standard tier, with the stronger backbone, SAM, Fish, DeepCORAL and GroupDRO are not flagged. Their largest edge

Table S7

Barlow Twins fine-tunes with the inherited ImageNet ResNet-50 learning rate and with their own tuned rate: mean ID-validation AUROC and mean off-site (test) AUROC at the ID-selected checkpoint, over folds and seeds 0–2.

Arm	Cohort	ID-validation		Off-site	
		inherited	tuned	inherited	tuned
BT ERM	Camelyon17	0.968	0.983	0.937	0.954
	Canine cSCC	0.825	0.925	0.776	0.885
	MIDOG 2021	0.865	0.894	0.829	0.885
	MIDOG++	0.775	0.924	0.715	0.891
BT + H&E jitter	Camelyon17	0.892	0.981	0.887	0.957
	Canine cSCC	0.759	0.920	0.782	0.913
	MIDOG 2021	0.842	0.895	0.817	0.891
	MIDOG++	0.766	0.927	0.687	0.896

Table S8

Standard-tier learning-rate selection: number of folds (of 20) that select each rate on the initial grid (to 10^{-2}), after the first extension (to 3×10^{-2}) and on the final grid (to 10^{-1}), with the final picks per cohort. Div.: diverged runs at that rate (of 20). Seed 0, ID-validation AUROC. – : rate not in the grid.

Backbone	Rate	Shorter grids		Final grid					Div.
		to 10^{-2}	to 3×10^{-2}	C17	Canine	MIDOG21	MIDOG++	All	
ResNet-50 (SGD)	10^{-3}	0	0	0	0	0	0	0	0
	3×10^{-3}	1	0	0	0	0	0	0	0
	10^{-2}	19	4	0	1	2	0	3	0
	3×10^{-2}	–	16	4	2	1	5	12	0
	10^{-1}	–	–	1	2	0	2	5	0
DINO ViT-S/16 (AdamW)	10^{-5}			1	0	2	0	3	0
	3×10^{-5}			3	4	1	7	15	0
	10^{-4}			1	1	0	0	2	0

counts are 10 of 20 folds, for Fish $\epsilon = 0.5$ and DeepCORAL $\lambda = 10$. IRMv1 $\lambda = 1$ (18) and the two-value axes of LISA, MixStyle and H&E jitter are flagged. On the un-normalized MIDOG 2021 cohort, the per-fold selections match those of the primary MIDOG 2021 cohort on all 3 folds for IRMv1 and H&E jitter, on 2 of 3 for GroupDRO, SAM, Fish, MixStyle and LISA, and on 1 of 3 for DeepCORAL.

Diverged runs Every diverged tuning run is an IRMv1 run: 29 of the 598 compact-tier runs and 32 of the 680 standard-tier runs. Divergence was detected at a checkpoint, when the training loss or a prediction became non-finite (Section 4.3). The IRMv1 penalty weight is 1 for the first 300 steps and λ afterwards. At the compact tier, $\lambda = 100$ diverged on all 20 primary folds and on the 3 un-normalized MIDOG 2021 folds. $\lambda = 10$ diverged on 6 of 20 (Camelyon17 4, MIDOG 2021 1, MIDOG++ 1), and $\lambda = 1$ never diverged. Divergence was detected at step 360, the first checkpoint after the switch, in 27 of the 29 runs and at step 480 in the other 2. Divergence therefore enforces part of the preference for $\lambda = 1$, but not all of it. On the 14 primary folds where $\lambda = 10$ completed, $\lambda = 1$ still had the higher ID-validation AUROC on 13. At the standard tier, $\lambda = 100$ diverged on all 20 folds, $\lambda = 10$ on 11 and $\lambda = 1$ on 1. The divergences were detected between steps 240 and 600 (240: 3 runs; 360: 23; 480: 5; 600: 1). They became more frequent at larger fold learning rates: 3 of 9 runs at 10^{-2} , 20 of 36 at 3×10^{-2} and 9 of 15 at 10^{-1} . On the 9 folds where $\lambda = 10$ completed, $\lambda = 1$ had the higher ID-validation AUROC on 7. On MIDOG++ fold 4 (learning rate 10^{-1}), all three IRMv1 configurations diverged at step 240, before the penalty weight changes. Their runs are therefore identical, with ID-validation AUROC 0.643 at their selected checkpoint (step 120). The fold falls back to $\lambda = 1$ by the tie-break and is the only fold of either tier in which every configuration diverged. No other arm had a diverged tuning run.

Table S9

Per-fold learning-rate picks at the standard tier (fold t tests domain t). ResNet-50 rates are given as decimals and ViT rates in units of 10^{-5} . The rows for the shorter ResNet-50 grids restrict each fold's final tuning records to the rates of that grid.

Cohort	Selection	Fold 0	1	2	3	4	5	6
Camelyon17	ResNet-50, grid to 10^{-2}	0.01	0.01	0.003	0.01	0.01		
	ResNet-50, grid to 3×10^{-2}	0.03	0.03	0.03	0.01	0.03		
	ResNet-50, final	0.03	0.03	0.03	0.1	0.03		
	ViT-S/16	1	3	10	3	3		
Canine cSCC	ResNet-50, grid to 10^{-2}	0.01	0.01	0.01	0.01	0.01		
	ResNet-50, grid to 3×10^{-2}	0.01	0.03	0.03	0.03	0.03		
	ResNet-50, final	0.01	0.1	0.03	0.1	0.03		
	ViT-S/16	3	10	3	3	3		
MIDOG 2021	ResNet-50, grid to 10^{-2}	0.01	0.01	0.01				
	ResNet-50, grid to 3×10^{-2}	0.03	0.01	0.01				
	ResNet-50, final	0.03	0.01	0.01				
	ViT-S/16	3	1	1				
MIDOG++	ResNet-50, grid to 10^{-2}	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	ResNet-50, grid to 3×10^{-2}	0.03	0.03	0.03	0.03	0.03	0.03	0.03
	ResNet-50, final	0.03	0.03	0.1	0.03	0.1	0.03	0.03
	ViT-S/16	3	3	3	3	3	3	3

Table S10

Compact-tier selections: number of folds that select each grid value (seed 0, ID-validation AUROC, learning rate 0.02). C17, Canine, MIDOG21, MIDOG++: 5, 5, 3 and 7 folds; All: the 20 primary folds. Div.: diverged tuning runs of that value (of 20). MIDOG21u: the un-normalized MIDOG 2021 sensitivity cohort (3 folds), with its diverged runs (of 3). Edge flag: an edge value of a grid axis selected on more than half of the 20 primary folds, with its count.

Arm	Value	C17	Canine	MIDOG21	MIDOG++	All	Div.	MIDOG21u		Edge flag
								Sel.	Div.	
GroupDRO	$\eta=10^{-3}$	3	2	0	5	10	0	0	0	none
	$\eta=10^{-2}$	1	2	2	2	7	0	1	0	
	$\eta=10^{-1}$	1	1	1	0	3	0	2	0	
IRMv1	$\lambda=1$	4	5	3	7	19	0	3	0	$\lambda=1$ (19)
	$\lambda=10$	1	0	0	0	1	6	0	0	
	$\lambda=100$	0	0	0	0	0	20	0	3	
DeepCORAL	$\lambda=0.1$	2	1	1	0	4	0	1	0	none
	$\lambda=1$	1	3	2	6	12	0	2	0	
	$\lambda=10$	2	1	0	1	4	0	0	0	
SAM	$\rho=0.02$	3	5	2	7	17	0	3	0	$\rho=0.02$ (17)
	$\rho=0.05$	2	0	1	0	3	0	0	0	
	$\rho=0.1$	0	0	0	0	0	0	0	0	
Fish	$\epsilon=0.01$	2	0	0	0	2	0	0	0	$\epsilon=0.5$ (13)
	$\epsilon=0.05$	1	0	0	0	1	0	0	0	
	$\epsilon=0.1$	2	1	0	1	4	0	1	0	
	$\epsilon=0.5$	0	4	3	6	13	0	2	0	
MixStyle	$p=0.5, \alpha=0.1$	1	0	1	0	2	0	2	0	$\alpha=0.3$ (13)
	$p=0.5, \alpha=0.3$	1	0	1	6	8	0	1	0	
	$p=1, \alpha=0.1$	2	2	1	0	5	0	0	0	
	$p=1, \alpha=0.3$	1	3	0	1	5	0	0	0	
LISA	$\alpha=0.5, p_{\text{sel}}=0.5$	2	2	3	2	9	0	2	0	$\alpha=0.5$ (16); $p_{\text{sel}}=0.5$ (11)
	$\alpha=0.5, p_{\text{sel}}=1$	1	1	0	5	7	0	1	0	
	$\alpha=2, p_{\text{sel}}=0.5$	1	1	0	0	2	0	0	0	
	$\alpha=2, p_{\text{sel}}=1$	1	1	0	0	2	0	0	0	
H&E jitter	$s=0.2$	4	5	3	7	19	0	3	0	$s=0.2$ (19)
	$s=0.4$	1	0	0	0	1	0	0	0	

Table S11

Standard-tier (ResNet-50) selections at each fold's selected learning rate (Table S9). Columns as in Table S10. ^aIncludes MIDOG++ fold 4, where all three IRMv1 configurations diverged and $\lambda = 1$ is kept by the tie-break.

Arm	Value	C17	Canine	MIDOG21	MIDOG++	All	Div.	Edge flag
GroupDRO	$\eta=10^{-3}$	0	2	1	4	7	0	none
	$\eta=10^{-2}$	3	3	1	3	10	0	
	$\eta=10^{-1}$	2	0	1	0	3	0	
IRMv1	$\lambda=1$	5	5	1	7 ^a	18 ^a	1	$\lambda=1$ (18)
	$\lambda=10$	0	0	2	0	2	11	
	$\lambda=100$	0	0	0	0	0	20	
DeepCORAL	$\lambda=0.1$	1	2	1	2	6	0	none
	$\lambda=1$	2	1	0	1	4	0	
	$\lambda=10$	2	2	2	4	10	0	
SAM	$\rho=0.02$	1	2	1	3	7	0	none
	$\rho=0.05$	2	2	1	2	7	0	
	$\rho=0.1$	2	1	1	2	6	0	
Fish	$\epsilon=0.01$	1	1	0	1	3	0	none
	$\epsilon=0.05$	0	1	0	1	2	0	
	$\epsilon=0.1$	1	2	1	1	5	0	
	$\epsilon=0.5$	3	1	2	4	10	0	
MixStyle	$p=0.5, \alpha=0.1$	2	3	0	5	10	0	$p=0.5$ (17); $\alpha=0.1$ (12)
	$p=0.5, \alpha=0.3$	2	2	1	2	7	0	
	$p=1, \alpha=0.1$	1	0	1	0	2	0	
	$p=1, \alpha=0.3$	0	0	1	0	1	0	
LISA	$\alpha=0.5, p_{\text{sel}}=0.5$	1	0	1	2	4	0	$\alpha=0.5$ (19); $p_{\text{sel}}=1$ (16)
	$\alpha=0.5, p_{\text{sel}}=1$	3	5	2	5	15	0	
	$\alpha=2, p_{\text{sel}}=0.5$	0	0	0	0	0	0	
	$\alpha=2, p_{\text{sel}}=1$	1	0	0	0	1	0	
H&E jitter	$s=0.2$	3	4	2	6	15	0	$s=0.2$ (15)
	$s=0.4$	2	1	1	1	5	0	

Table S12

Worst-domain AUROC (ID-validation selection) of the stain-normalization arms, with ERM and H&E jitter for reference, and the off-site AUROC on Camelyon17 center 4 (C17 c4). Means over 5 seeds (compact tier) and 3 seeds (standard tier).
[†]Transductive.

Arm	Compact tier					Standard tier				
	C17 c4	C17	Canine	MIDOG21	MIDOG++	C17 c4	C17	Canine	MIDOG21	MIDOG++
ERM	0.698	0.698	0.551	0.765	0.766	0.977	0.895	0.534	0.847	0.870
H&E jitter	0.917	0.904	0.674	0.864	0.809	0.985	0.958	0.850	0.875	0.880
Macenko (slide) [†]	0.952	0.904	0.698	0.870	0.828	0.976	0.915	0.819	0.860	0.873
TIA-style [†]	0.982	0.889	0.730	0.864	0.810	0.960	0.934	0.823	0.871	0.877
Macenko (tile)	0.524	0.524	0.671	0.844	0.816	0.557	0.557	0.783	0.846	0.877

S9. Stain-normalization granularity

Per-slide and per-tile estimation Our Macenko arms estimate one stain matrix per slide and scanner (Section 4.4). The per-tile variant runs the same estimator on every tile separately, with the non-negativity clamp, and maps it to a reference fitted on 256 training tiles. A tile with less than 5% tissue keeps the reference matrix. Both variants use no labels. Both were run with the full protocol: 5 seeds at the compact tier, 3 at the standard tier, all 20 folds.

Result Table S12 gives the worst-domain AUROC of each variant. The two differ mainly on Camelyon17 center 4. There, per-tile normalization gives an off-site AUROC of 0.524 at the compact tier and 0.557 at the standard tier. With per-slide normalization the scores are 0.952 and 0.976. The model is not the cause: standard-tier ERM without normalization scores 0.977 on center 4. On the other four Camelyon17 centers, the per-tile variant is within 0.018 of the per-slide one at the compact tier and equal or better at the standard tier, by up to 0.034. On canine cSCC, per-slide normalization is higher by 0.027 (compact) and 0.036 (standard). On the two MIDOG cohorts the variants differ by at most 0.026. On Camelyon17 the choice of granularity alone moves the arm from rank 4 of 20 to last at the compact tier, and from rank 10 of 21 to last at the standard tier. The failing domain has the palest tissue: center-4 tiles are 27–34% tissue pixels, against 58–83% in the other centers. A per-tile estimate rests on the 1,100–1,400 tissue pixels of a 64 px tile, usually of a single tissue type.

Negative concentrations The earlier version of this pipeline used per-tile estimation with unconstrained least-squares concentrations, as in Macenko’s reference code. In a 64 px tile the estimated hematoxylin and eosin vectors are often nearly collinear (cosine 0.95–0.98). Eosin-rich pixels then receive large negative hematoxylin values and come out saturated pink after reconstruction. On center 4 this artifact affected 90% of tumor tiles but 54% of normal tiles. The off-site AUROC on center 4 was 0.42 (5 seeds, compact tier), below every arm of Table S12. Clamping the concentrations at zero removes the artifact but not the unreliable per-tile stain matrix. The per-slide estimate removes both. The earlier records are kept apart and are not used in any table.

Dependence on the tile set The per-slide estimate of a test slide uses that slide’s own unlabeled tiles. In Camelyon17 those tiles are the benchmark’s patches, sampled from annotated regions and class-balanced. The share of tumor tiles per test slide therefore ranges from 0 to 0.93. To bound the effect of this makeup, we fixed the trained weights of five compact-tier runs on center 4 (seeds 0–4, final configuration) and re-estimated each test slide’s stain matrix from different subsets of its tiles. Labels were used only to form these diagnostic subsets. The mean off-site AUROC was 0.945 with all tiles, 0.948 with only negative tiles and 0.934 with only positive tiles: a spread of 0.014. With no per-slide estimate, using only the reference, it fell to 0.560. For TIA-style the corresponding values were 0.981, 0.972, 0.978 and 0.897. The trained models therefore rely on the per-slide normalization, but not much on which tiles of the slide feed it. A label-independent tissue sample of each whole-slide image would match deployment more closely. It is not available, because the cohorts are distributed as patches.

S10. External example: the variance decomposition on DomainBed

To give the ICC of Section 3.3 a reference point outside pathology, we apply the same estimators to the published per-environment results of DomainBed (Gulrajani and Lopez-Paz, 2021). DomainBed also evaluates by leave-one-environment-out, so each environment is the test environment of one fold, as in our design.

Source We cite the ICLR 2021 version of the paper. The per-environment tables, however, are taken from the appendix “Domain generalization accuracies per algorithm, dataset, and domain” of the first arXiv e-print (arXiv:2007.01434v1). It is the only version whose source arXiv provides, and it reports nine algorithms: ERM, IRM, DRO, Mixup, MLDG, CORAL, MMD, DANN and C-DANN. The appendix has one table for each of seven datasets and three model-selection rules, with 837 cells of the form mean $\pm$ standard error (SE). A script parses all of them and stores each with its source line. Accuracies are in percentage points (pp), rounded to 0.1. In the Colored MNIST table for training-domain validation the two adversarial algorithms are named ADA and CondADA, while every other table names them DANN and C-DANN. We treat them as the same algorithms. The means of the ADA and CondADA rows over the three environments are 51.5 and 51.9, which are the Colored MNIST entries of DANN and C-DANN in the main results table of the same e-print.

From standard errors to standard deviations Each cell is the mean over three trials, and each trial repeats the whole study with new random hyperparameters, weight initializations and dataset splits. The paper does not give the formula for its SE. We take it from the first public release of the DomainBed code (commit 8ff97a7), in which `collect_results.format_mean` computes the population SD (divisor n) of the trial accuracies divided by $\sqrt{n}$. We assume that the paper’s tables were produced with this function. The sample SD (divisor $n - 1$) of a cell with $n = 3$ trials is then

$$s = \text{SE} \cdot \frac{n}{\sqrt{n-1}}. \quad (\text{S1})$$

The textbook convention $s = \text{SE} \cdot \sqrt{n}$ gives SDs smaller by a factor $\sqrt{(n-1)/n}$, and hence larger ICCs. We report it only as a sensitivity check. An SE printed as 0.0 is below 0.05 and is used as zero.

Model For each algorithm and dataset under training-domain validation (the first rule in DomainBed’s main results table, and our primary rule), we fit model (5) to the test accuracy y_{ts} of held-out environment t in trial s . This gives 63 fits, with K between 3 and 6 environments and $n = 3$ trials each. The published summaries are a mean and an SE per cell, so trials can only be treated as nested within environments, and no crossed trial effect can be fitted. All estimators of Section 3.3 depend on the data only through the cell means and the within-cell sums of squares. We therefore reconstruct three pseudo-observations per cell that reproduce the published mean and the SD of (S1). The analysis of these pseudo-data is exact. With $n = 3$ trials in every cell, the data are balanced and the ICC interval (S2) is exact. As a check, we also compute the grid posterior (S3) with half-Cauchy (scale 10 pp) and half-normal (scale 25 pp) priors on both SDs.

Results Table S13 and Fig. S15 summarize the 63 fits. No dataset is run-dominated. The median ICC over the 63 fits is 0.98, and the ICC exceeds 0.8 in all 54 fits outside TerraIncognita. Colored MNIST, DomainNet, Office-Home and VLCS are the most strongly environment-dominated (median ICC above 0.98, median interval width below 0.08). On Colored MNIST, every algorithm reaches 10.0–10.5% accuracy on the environment labeled 0.9 in the source table and 71.4–73.3% on the other two, and the median $\hat{\sigma}_{\text{env}}$ is 36.1 pp against a median $\hat{\sigma}_{\text{run}}$ of 0.5 pp. Rotated MNIST has the smallest spread between environments (median 1.5 pp), but its between-trial SD is smaller still (0.3 pp). This dataset also has the most SEs printed as 0.0 (23 of 54 cells), so rounding probably understates its run variance and overstates its ICC. On PACS the median ICC is 0.96, but two algorithms (IRM and C-DANN) have intervals that reach below 0.5. TerraIncognita is the exception. Its median ICC is 0.77, its median $\hat{\sigma}_{\text{run}}$ (4.0 pp) is about half its median $\hat{\sigma}_{\text{env}}$ (7.5 pp), and six of its nine algorithms have intervals with lower limits below 0.5. For MMD, the ICC is 0.51 with interval [0.00, 0.95], and the F test of no environment effect gives $p = 0.048$. ERM has ICC 0.93 [0.67, 0.99] there. With four environments, the exact interval has 3 numerator degrees of freedom, and on TerraIncognita the median width is 0.72. The Bayesian medians are close to the ANOVA estimates (median 0.76 on TerraIncognita).

These conclusions do not depend on the SE convention. With the textbook convention $s = \text{SE}\sqrt{n}$, the median ICC on TerraIncognita is 0.84 and five fits, rather than eight, have lower limits below 0.5. They do depend on the selection

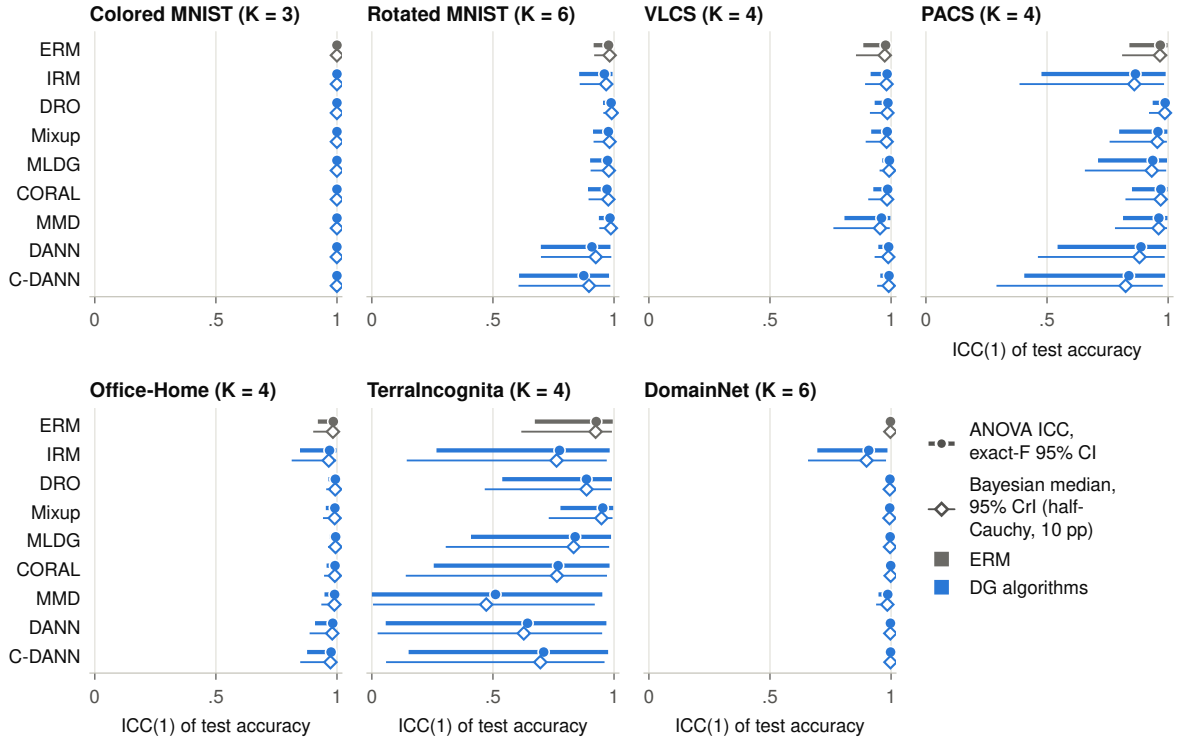
Table S13

ICC(1) of test accuracy in DomainBed under training-domain validation, summarized over the nine algorithms of each dataset (arXiv:2007.01434v1, appendix tables; three trials per cell; SD from SE by (S1)). ICC: median and range over algorithms, with the algorithm with the lowest ICC. CI width: median and range of the width of the exact 95% interval (S2). Lower < 0.5: number of algorithms whose interval has a lower limit below 0.5. $\hat{\sigma}_{\text{env}}$ and $\hat{\sigma}_{\text{run}}$: medians over algorithms of the between-environment and between-trial SDs, in pp. SE = 0.0: cells printed with an SE of 0.0, out of the dataset's cells under this rule.

Dataset	K	SE = 0.0	ICC		Lowest	CI width median (range)	Lower < 0.5	$\hat{\sigma}_{\text{env}} / \hat{\sigma}_{\text{run}}$
			median	range				
Colored MNIST	3	5/27	1.00	1.00–1.00	DANN	< 0.01 (< 0.01–< 0.01)	0	36.1 / 0.5
Rotated MNIST	6	23/54	0.97	0.87–0.99	C-DANN	0.09 (0.04–0.37)	0	1.5 / 0.3
VLCS	4	0/36	0.99	0.96–0.99	MMD	0.07 (0.04–0.19)	0	14.4 / 1.8
PACS	4	2/36	0.96	0.84–0.99	C-DANN	0.20 (0.06–0.58)	2	9.5 / 2.1
Office-Home	4	0/36	0.99	0.97–0.99	IRM	0.05 (0.03–0.15)	0	11.5 / 1.1
TerraIncognita	4	0/36	0.77	0.51–0.95	MMD	0.72 (0.22–0.95)	6	7.5 / 4.0
DomainNet	6	2/54	1.00	0.91–1.00	IRM	0.01 (0.01–0.29)	0	19.3 / 1.2

rule. Under leave-one-domain-out validation, 17 of the 63 fits have lower limits below 0.5, compared with 8 under training-domain validation and 1 under test-domain (oracle) validation. One example is DANN on Rotated MNIST, whose three trials disagree so strongly ($\hat{\sigma}_{\text{run}} = 15.0$ pp) that its ICC is 0.00.

Implications for single-environment reporting Under training-domain validation, most of the variance of a single DomainBed run is due to which environment is held out. On six of the seven datasets, the median over algorithms of the ratio $\hat{\sigma}_{\text{env}} / \hat{\sigma}_{\text{run}}$ is between 4.8 and 73.5, and on TerraIncognita it is 1.9. A result on one held-out environment therefore describes mainly that environment. On these six datasets the spread between environments is several times the trial-to-trial SD, so a single environment cannot stand in for the dataset. This is a reason to evaluate on every environment, as DomainBed does, and to report the worst environment and the spread over environments together with the mean. On TerraIncognita, the trial-to-trial SD is a large fraction of the spread between environments, so a single trial says little about a given environment, even though each trial already includes a fresh hyperparameter search. Replicating trials is necessary there, not optional. A high ICC does not show that one algorithm beats another in every environment. That question concerns the paired difference (8), whose variance excludes the shared environment effect. DomainBed's trials also differ from our seeds. A DomainBed trial redraws the hyperparameters and the data split, whereas our seeds keep both fixed within a fold (Section 3.1). The two ICCs are therefore not on the same scale. The comparison shows only that the ICC can be computed from what a benchmark already publishes, and that its value, and how precisely it is known, varies with the dataset, the number of environments and the selection rule.



DomainBed (Gulrajani & Lopez-Paz 2021), per-environment tables of the e-print appendix, training-domain validation, 3 trials; SD from SE as in DomainBed's collect_results. Colored MNIST rows 'ADA' / 'CondADA' of the source table are DANN / C-DANN.

Figure S15: ICC(1) of test accuracy for the nine DomainBed algorithms on seven datasets under training-domain validation (arXiv:2007.01434v1, appendix tables; three trials per cell; SD from SE by (S1)). Filled circles and thick bars: ANOVA estimate and exact 95% interval (S2). Open diamonds and thin bars: posterior median and equal-tailed 95% credible interval under half-Cauchy priors (scale 10 pp). ERM is shown in gray, the domain-generalization algorithms in blue. The Colored MNIST rows ADA and CondADA of the source table are shown as DANN and C-DANN.

S11. Additional protocol details

S11.1. Interval estimates for the variance components

With $F = \text{MSB}/\text{MSW}$, the exact 95% interval for ICC(1) (Searle et al., 1992; McGraw and Wong, 1996) is

$$\left[\frac{F_L - 1}{F_L + n_0 - 1}, \frac{F_U - 1}{F_U + n_0 - 1} \right], \quad F_L = \frac{F}{F_{0.975}(K-1, N-K)}, \quad F_U = \frac{F}{F_{0.025}(K-1, N-K)}, \quad (\text{S2})$$

clipped to $[0, 1]$ (exact for balanced data, approximate otherwise). σ_R receives the exact χ^2_{N-K} interval and σ_D the modified large-sample interval (Burdick and Graybill, 1992, §3.3).

S11.2. Bayesian posterior

With $K \leq 7$, $\hat{\sigma}_D$ is often at or near zero. We therefore also report a grid posterior,

$$p(\sigma_D, \sigma_R \mid \mathbf{y}) \propto L_R(\sigma_D, \sigma_R) \pi(\sigma_D) \pi(\sigma_R), \quad (\text{S3})$$

where L_R is the restricted likelihood (μ integrated out under a flat prior). π is either a half-normal prior, $\pi(\sigma) \propto \exp\{-\sigma^2/(2A^2)\}$, or a half-Cauchy prior, $\pi(\sigma) \propto (1 + \sigma^2/A^2)^{-1}$ (Gelman, 2006). Both are independent on each SD with scale $A = 0.1$ AUROC for both σ_D and σ_R . The half-normal places 95% of its mass below 0.20; the half-Cauchy has heavy tails. The posterior is computed on a 400×300 grid (zero plus a geometric grid for σ_D ; a geometric grid spanning the χ^2 range of the within-domain sum of squares for σ_R) with trapezoid weights. We report posterior medians and equal-tailed 95% credible intervals for σ_D , σ_R and the ICC. When REML is at the boundary or the ANOVA estimate is truncated, we report the one-sided interval $[0, q_{0.95}]$, because an equal-tailed interval cannot contain zero.

S11.3. Response shape

Whether gains are larger where ERM is weak is assessed with Oldham's correlation $r = \text{corr}(M_t - E_t, (M_t + E_t)/2)$ between the per-domain difference and mean of the arm (M_t) and ERM (E_t) (Oldham, 1962). Unlike a regression of gain on baseline, it is not biased by regression to the mean. $r = 0$ if and only if $\text{Var}(M) = \text{Var}(E)$, which is tested by the Pitman–Morgan statistic $t = r\sqrt{K-2}/\sqrt{1-r^2}$ on $K-2$ degrees of freedom (Pitman, 1939; Morgan, 1939). We also give an exact sign-flip permutation p -value over all 2^K swaps of (M_t, E_t) . The results are in Section S7.

S12. Additional details of the cohorts and arms

S12.1. Cohorts

Camelyon17-WILDS The cohort consists of 96×96 px lymph-node patches labeled tumor or normal from five hospitals (Koh et al., 2021; Bandi et al., 2019). In fold t , the test split is a class-balanced sample of 25,000 patches of hospital t , the OOD-validation split 12,000 patches of hospital $(t-1) \bmod 5$, and the training split 30,000 class-balanced patches of the other three hospitals. ID validation (12,000 patches) holds out $\max(1, \lfloor n_c/5 \rfloor)$ of the n_c patients of every training hospital c . The patients are redrawn with a seeded generator until the held-out set has at least 1,000 patches of each class and at most 40% of the hospital's tumor patches, because patients contribute between 5 and 56,666 tumor patches. The 64 px cache is the central crop of the native patch.

Canine cutaneous squamous cell carcinoma Each of 44 slides was digitized with five scanners (Wilm et al., 2023): cs2 (Aperio ScanScope CS2), gt450 (Aperio GT450), p1000 (3DHistech Pannoramic 1000), nz20 (Hamamatsu NanoZoomer 2.0-HT) and nz210 (NanoZoomer S210). A domain is a scanner. The slides are randomly split into five groups. Fold t tests scanner t on slide group t , validates on scanner $(t-1) \bmod 5$ with slide group $(t-1) \bmod 5$, and trains on the remaining three scanners and slide groups. A test domain is therefore a new scanner and a set of slides that is absent from training under every scanner. Four training slides (15%) form the ID-validation split. Per image, about 420 candidate centers are drawn, half in carcinoma polygons and half in non-tumor tissue polygons, allocated in proportion to polygon area. Negative centers that fall inside any tumor polygon are dropped. A patch is kept if it lies inside the image at both sizes and its 64 px crop passes an intensity filter (mean ≤ 225 , SD ≥ 10), and duplicate centers are removed. Magnification is normalized to cs2: the factor of each scanner is the median over slides of the square root of its annotation-area ratio to cs2. Crops of 64/65/71/73/62 px (cs2, p1000, nz20, nz210, gt450) are resized bilinearly to 64 px, and crops of 96/98/107/110/92 px to 96 px. Factors recomputed without each fold's test and validation slides give the same crop sizes.

MIDOG 2021 The training set of the MIDOG 2021 challenge (Aubreville et al., 2023a) contains 150 annotated human breast cancer images from one laboratory, 50 from each of three scanners (Hamamatsu NanoZoomer XR, NanoZoomer S360 and Aperio ScanScope CS2). A domain is a scanner ($K = 3$), and there is no OOD-validation domain. Patches are centered on the annotated bounding boxes of mitotic figures and non-mitotic look-alikes. ID validation holds out 15% of the images of every training scanner (eight per scanner), fixed per scanner. The TIFF resolutions are 0.2263–0.2274 (XR), 0.2298 (S360) and 0.2533 (CS2) $\mu\text{m}/\text{px}$, a difference of about 11%. The primary cache therefore normalizes every image to 0.25 $\mu\text{m}/\text{px}$: it crops round($P \cdot 0.25/\text{mpp}$) px (63–71 and 95–106 px) and resizes bilinearly to $P \in \{64, 96\}$. The un-normalized cache keeps the magnification difference as part of the shift, as in the challenge. It is run at the compact tier as a sensitivity analysis.

MIDOG++ MIDOG++ (Aubreville et al., 2023b) provides 503 annotated 2 mm² regions of interest from seven tumor types in two species (Table S14). Its 150 breast cancer images have pixel data identical to the MIDOG 2021 images. A domain is a tumor type ($K = 7$), and laboratory, scanner and species vary with it. Task, patch centering, scale normalization and ID validation follow MIDOG 2021. For the 3DHistech images, whose x and y resolutions differ by 0.4%, the geometric mean is used.

Mirror padding MIDOG objects closer to the image border than half a crop are cut from a reflect-padded copy of the image. The object therefore stays centered, and its 64 and 96 px patches are concentric. At 96 px this affects 111 of 4,435 MIDOG 2021 patches and 617 of 26,286 MIDOG++ patches, at similar rates in both classes (MIDOG++: 2.43% of positives, 2.28% of negatives). The padding of each row is stored for a sensitivity analysis that excludes these patches.

Unlabeled pools The pool used for self-supervised pre-training comes only from the training domains and excludes ID-validation groups, the validation domain and the test domain. The arms that use it therefore remain domain generalization methods and do not adapt to the test domain. On Camelyon17 the pool consists of 40,000 patches of non-ID-validation training patients that are not in the training split. On the other cohorts it consists of random tissue crops (mean < 220 , SD > 12) from non-ID-validation training images. The crops are drawn with image-specific seeds, at most 800 per canine image and 300 per MIDOG image, and subsampled to 40,000. MIDOG 2021 yields only 25,200.

Table S14

MIDOG++ domains (scanners and laboratories from Table 1 of Aubreville et al., 2023b). Counts are the annotated mitotic figures and non-mitotic look-alikes used in this study; resolution is from the TIFF tags. Fold t tests domain t and validates on domain $(t - 1) \bmod 7$.

Tumor type	Species	Images	Mitotic	Look-alike	Scanner	Laboratory	$\mu\text{m}/\text{px}$
Mast cell tumor	canine	50	2,327	1,366	Aperio ScanScope CS2	FU Berlin	0.2533
Lung cancer	canine	44	855	951	3DHistech	VMU Vienna	0.2487
Lymphosarcoma	canine	55	3,959	4,257	Pannoramic Scan II	VMU Vienna	0.2487
Soft tissue sarcoma	canine	100	1,286	2,375	3DHistech	AMC New York, VMU Vienna	0.2487
Breast cancer	human	150	1,721	2,714	Pannoramic Scan II	UMC Utrecht	0.2263–0.2533
					Hamamatsu XR, S360; Aperio CS2		
Melanoma	human	49	1,150	925	Hamamatsu XR	UMC Utrecht	0.2263–0.2274
Neuroendocrine tumor	human	55	639	1,761	Hamamatsu XR	UMC Utrecht	0.2263–0.2274
Total		503	11,937	14,349			

S12.2. Self-supervision controls and stain normalization

Controls for self-supervision The augmentation-only control applies SimCLR’s view augmentation to labeled batches without the contrastive loss. It separates the effect of the augmentations from that of pre-training. As in the SimCLR views, crop and flip parameters are drawn per batch and color factors per image. The compute-matched control gives ERM as many images through the encoder as SimCLR receives in pre-training and fine-tuning together. Pre-training runs 6 epochs of $\lfloor |U|/256 \rfloor$ steps on two views of 256 images, so $I_{\text{SSL}} = 479,232$ for $|U| = 40,000$ and 301,056 for MIDOG 2021. SimCLR is pre-trained separately for every fold and seed, with SGD (momentum 0.9, weight decay 10^{-4}), a cosine schedule and projection heads of 256–256–128 (C) or 2048–2048–128 (S). It is then fine-tuned with the ERM recipe.

Stain normalization As in common practice, Macenko’s stain matrix (Macenko et al., 2009) is estimated once per slide. A group is a slide imaged on one scanner: a Camelyon17 slide, a MIDOG image, a canine slide on one of its three scanners, or a PatchCamelyon whole-slide image. For each group, up to 256 tiles are drawn in a content-hashed order that does not depend on row order. Their tissue pixels (optical density ≥ 0.15 in every channel; at most 200,000, subsampled with a fixed seed) are pooled. Macenko’s estimator is then applied once: the plane of the two leading eigenvectors, then the 1st and 99th angular percentiles. The two directions are labeled hematoxylin and eosin by matching them to the reference. Concentrations are the least-squares solution clamped at zero, because without the clamp eosin-rich pixels received large negative hematoxylin values on pale tiles. The group’s maximal concentrations are the per-stain 99th percentiles over its pooled pixels. Every tile is rescaled to the reference maxima and reconstructed with the reference matrix. A group with too little tissue keeps the reference. The reference is fitted per fold with the same estimator on every training group: its stain matrix is the elementwise median of the aligned group matrices, and its maxima are the median group maxima. No labels are used. The method is transductive, however: each test slide is normalized from its own tiles, which are the benchmark’s patches from annotated regions rather than a label-independent sample of the slide. Section S9 reports how much the estimate depends on that tile set, and a per-tile variant that estimates the stain matrix on every tile separately. Our H&E jitter perturbs RGB color; it is not an augmentation in HED space.

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